Can AI Do Financial Research?
LLM-Guided Hypothesis Discovery in Asset Pricing*

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Abstract

We study whether an AI research system can autonomously execute the hypothesis-discovery loop in empirical asset pricing within a human-designed laboratory. We place specialized large language model agents in an environment with a symbolic language of interpretable accounting formulas and a fixed empirical evaluation pipeline. Over successive generations, the agents propose, critique, test, and revise hypotheses about interactions among economic themes such as profitability, investment, valuation, and quality, using standardized feedback from prior generations. Across three separately initialized primary runs, the system evaluates approximately 1,600 hypotheses per run; 59.1%–62.3% clear a conventional significance screen in the predicted direction. We apply multiple-testing corrections and jointly compare related formulas, retaining 52–68 candidates per run with distinct predictive content. We then test their stability over time, robustness across factor models, and incremental information relative to 209 published anomalies, identifying a small and robust set that survives the full pipeline. While AI makes hypothesis generation inexpensive and may worsen asset pricing's longstanding lack of search transparency, our paper introduces a human-designed architecture in which specialized AI agents search autonomously, every hypothesis and feedback-driven revision is recorded and auditable, and the resulting search universe is subjected to formal statistical inference.

Keywords: Asset pricing, AI agent, large language models, scientific discovery, hypothesis generation

JEL Classification: G12, G14, C45, C58

*All errors are our own. Code, data, and the complete discovery record are available at https://github.com/Chihche-Liew/signal-laboratory. The repository also provides an agent-readable record linking each claim to its supporting run artifacts and preserving both successful and failed hypotheses for cumulative learning by future AI research agents. The views expressed in this paper are solely those of the authors and do not necessarily reflect the views of Google or its affiliates.

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1 Introduction

Asset pricing has a hidden search denominator. The evidentiary weight of a reported return predictor depends not only on its t-statistic, but also on how many ideas, transformations, signs, and implementations were examined before it was selected. Yet published research generally reveals successful specifications, not the unsuccessful search that produced them. Harvey et al. (2016) show why conventional significance thresholds are inadequate under extensive multiple testing, while Harvey (2017) explains how unreported tests, publication incentives, and specification search weaken the credibility of financial evidence.

Recent work shows that LLMs can extract return-predictive information from text and automate substantial parts of financial research (Lopez-Lira and Tang, 2023; Novy-Marx and Velikov, 2026). Yet AI also sharpens the hidden-denominator problem. By lowering the cost of producing and evaluating hypotheses, it can scale specification search and greatly enlarge the hidden denominator. The same technology, however, offers a countervailing possibility. When AI operates inside an instrumented research environment, every proposal, revision, rejection, and empirical evaluation can be recorded. We therefore study whether an AI research system can autonomously generate, test, and refine new asset-pricing hypotheses while making the complete search path observable and incorporating the resulting search denominator into rigorous statistical inference.

At the center of the design is a symbolic expression language: a constrained grammar with 66 Compustat variables and 32 composable operators, including ratios, changes, rolling statistics, ranks, and industry adjustments. The agent uses the economic context of a pair of themes, such as profitability and investment or valuation and quality, to generate interpretable accounting formulas as candidate hypotheses that can be read as economic statements about firms. Within each theme-pair task, a proposer agent generates hypotheses using evidence archived from earlier generations, and a critic agent evaluates their economic logic, measurement, and distinctiveness before seeing their new empirical results. Flagged proposals return to the proposer for revision, after which deterministic code validates each formula and evaluates it through a fixed CRSP–Compustat pipeline using portfolio sorts and Fama–MacBeth regressions. The resulting empirical summaries enter the archive for the next generation. Human researchers define the hypothesis language, research workflow, sample, and empirical protocols in advance; the agents choose which economic ideas to pursue but cannot alter how those ideas are tested. This separation ensures that the evaluation standards are not influenced by the

discovery process.

Asset pricing provides a demanding and informative laboratory for this exercise. The domain combines a long historical panel from 1963 to 2024, well-established empirical protocols, and a rich literature on anomaly discovery, replication, and multiple testing (Harvey et al., 2016; Harvey, 2017; Harvey and Liu, 2020; Hou et al., 2020; Chen and Zimmermann, 2022). It is therefore a setting in which proposed hypotheses can be evaluated against well-defined standards rather than through ad hoc case studies. We organize eight accounting themes into all 28 cross-theme pairs. Each run contains ten generations for every pair and requests six proposals per generation, yielding approximately 1,600 empirical evaluations. We conduct three separately initialized primary runs and treat each as a separate search and testing universe. Following the investability concerns emphasized by Hou et al. (2020), we exclude financial firms and microcaps, and use NYSE breakpoints and value weights in the portfolio tests. Across the three runs, 59.1%–62.3% of evaluated hypotheses clear a conventional significance screen in the predicted direction. After multiple-testing corrections and joint comparisons among related formulas, 52–68 candidates per run retain both statistical support and incremental predictive content.

Our baseline evidence suggests that the framework supports productive and interpretable hypothesis discovery. We define the first-pass success rate as the share of evaluated hypotheses whose Fama–MacBeth t-statistic exceeds 1.96 in absolute value and whose estimated sign matches the direction predicted in advance. Across the three primary runs, this rate ranges from 59.1% to 62.3%. The generation-level results are consistent with learning from accumulated evidence: from Generation 1 onward, the median first-pass success rate rises from 43.4% to 74.3% by Generation 9, while the median t-statistic of the strongest correct-sign hypothesis increases from 5.5 in Generation 0 to between 7.2 and 7.7 in Generations 6–9. Similar improvement without critique points to the archive of prior results as the main source of learning across generations. Critique plays a different role: it broadens the range of economic mechanisms explored and increases the yield of hypotheses with distinct predictive content rather than simply raising the initial success rate.

The discovery trace makes this adaptation observable. Each hypothesis includes an economic mechanism, an executable expression, and a predicted sign, and later generations receive the complete archive of earlier outcomes. The critic flags roughly half of the hypotheses in each primary run, often identifying measurement problems, weak economic logic, or close variants

of ideas already examined. Revisions may repair how an economic idea is measured or redirect the search toward a different mechanism. We study these changes using observable records—proposal rationales, critic assessments, revisions, and empirical outcomes—rather than unobserved internal reasoning.

Repeated runs are important because LLM outputs are stochastic and feedback makes the subsequent search path dependent. We therefore conduct three separately initialized primary runs both to broaden coverage of the hypothesis space and to examine convergence as evidence of reproducibility. The runs produce remarkably similar aggregate discovery funnels and broadly agree on which economic theme pairs are most productive. Related economic mechanisms also recur across runs, even when expressed differently. Exact hypotheses, however, rarely overlap, and no final hypothesis appears in more than one run. Reproducibility therefore emerges more strongly in the structure of the search and the economic mechanisms it generates than in the identity of the final discoveries.

The main finding, however, is more disciplined than the baseline evidence. Because the agents revise hypotheses using outcomes computed on the same historical panel, the discovery loop is inherently an in-sample evolutionary search. We therefore place the inferential weight on a post-discovery validation framework rather than on the initial screen itself. This framework includes multiple-testing corrections (Harvey et al., 2016), skeptical Bayesian updating (Harvey, 2017), signed double-bootstrap analysis (Harvey and Liu, 2020), tests of stability and decay, factor-adjusted alphas under the Fama–French five-factor, Fama–French six-factor, and Hou–Xue–Zhang q-factor models (Fama and French, 2015; Hou et al., 2015), and conditional tests against 209 published predictors (Chen and Zimmermann, 2022). We also remove empirically near-duplicate hypotheses and use joint Fama–MacBeth regressions to retain only those that provide predictive information beyond other candidates generated for the same economic task.

The resulting attrition is substantial but informative. Of approximately 1,600 evaluated hypotheses per primary run, 949–1,001 clear the initial screen, 52–68 remain after multiple-testing corrections and joint comparisons, 37–45 remain stable without substantial decay, and 18–23 clear the alpha tests under all three factor models. Two hypotheses in each primary run survive the complete pipeline and retain incremental predictive content relative to published predictors. These results show why observing and testing the search denominator matters: many hypotheses appear successful in isolation, but only a small set survives the strictest standards.

Next, we examine the discovery traces behind several flagship hypotheses among the final survivors to understand how the agents developed and refined distinct economic mechanisms. The first, build versus buy knowledge capital, predicts that profitable firms relying on internal R&D may be undervalued because accounting expenses internally developed knowledge while capitalizing purchased intangibles. The second, funded research tilt, predicts that a shift from physical investment toward R&D is informative only when accompanied by gross-profit growth, which separates productive transformation from distress-driven cuts in capital spending. The third combines R&D-adjusted profitability with lean noncash working capital. Its weakest-link structure identifies firms whose conservatively reported profits are supported by operating discipline, rather than allowing strength on one dimension to offset weakness on the other. These discoveries turn familiar accounting inputs into new conditional hypotheses about when those inputs should predict returns.

The discovery traces show three corresponding forms of agent reasoning. In the first case, critique corrected a flawed measure, after which the agent extended the economic idea from investment flows to stocks of knowledge capital. In the second, critique redirected the agent from another profitability-and-investment combination toward a different question about resource reallocation and its operating payoff. In the third, criticism of a close variant of existing profitability signals led the agent to introduce R&D accounting conservatism and make working-capital discipline a binding condition. Together, these cases show that empirical feedback and critique can change not only how an idea is measured, but also the economic question being asked and the logical structure of the resulting hypothesis.

To our knowledge, this is the first study to combine autonomous LLM-guided signal discovery with modern multiple-testing corrections, alternative factor models, subsample and decay analyses, and spanning tests against published anomalies, while preserving a complete record of proposals, failures, and revisions. The paper makes four contributions. First, we introduce a finance-native discovery framework in which humans design and freeze the research environment while specialized AI agents autonomously generate, critique, and revise hypotheses. Second, our framework makes the search denominator observable by recording every evaluated hypothesis and incorporating the full search population into statistical inference, thereby addressing the multiple-testing concerns that AI's low-cost hypothesis generation can intensify. Third, we show that this architecture can discover interpretable return predictors with substantial economic content, including a small set that survives the most demanding tests. Fourth,

the recorded discovery traces provide direct evidence on how AI agents diagnose measurement problems, learn from earlier results, redirect unproductive searches, and develop new economic mechanisms. The paper therefore contributes both to the economics of the discovered predictors and to the design of transparent, AI-assisted financial research.

Our research produces a scientific record far richer than a conventional article or any individual reader can absorb. Beyond the small set of surviving predictors, the archive contains weak standalone signals, wrong-sign results that contradict their proposed economic mechanisms, and related signals whose apparent predictive content is absorbed by previously discovered factors or predictors. These outcomes are not merely discarded attempts. When hypotheses, predictions, and tests are specified and recorded consistently, well-documented failures can be as important to cumulative knowledge as successes: they reveal which mechanisms do not operate, which measurements fail to capture the intended idea, and where seemingly new evidence is subsumed by what is already known. Yet the prevailing publication model is necessarily organized around a small number of positive findings and provides no practical infrastructure for preserving and synthesizing such negative and boundary evidence at scale. We therefore complement the human-readable paper with a machine-readable archive that allows future AI research agents to learn from the full set of hypotheses and outcomes, not only the eventual winners. This pairing supports a new form of human–AI collaboration: humans provide interpretation and judgment, while AI preserves, organizes, and builds on a scientific record too large for any individual researcher to absorb.[1]

Two caveats discipline our interpretation. First, the search is adaptive and uses the same historical panel throughout, so our post-discovery tests provide search-adjusted inference rather than true out-of-sample validation. A true out-of-sample test would freeze the discovered hypotheses and all evaluation choices before new returns are realized, then evaluate them prospectively without further revision. Second, because LLM search is stochastic and path dependent, exact reproduction of the same search path and final hypotheses is generally infeasible. Our repeated runs therefore assess convergence in a broader economic sense—similar discovery fun-

[1]Our protocol follows recent computer science research on *agent-native research artifacts* (Liu et al., 2026), which organizes research so that both humans and future AI research agents can inspect, reproduce, and extend it. Rather than treating the paper as the complete scientific record, this approach links each claim to the experiments, code, and evidence that support it. It also preserves failed approaches and rejected alternatives as explicit parts of the discovery history, allowing future agents to learn from them rather than repeat them. A concise index helps an agent locate relevant material before retrieving the underlying details. Our implementation—including code, data, the complete discovery archive, and a machine-readable version of the paper's claims linked to their supporting run artifacts—is available at https://github.com/Chihche-Liew/signal-laboratory.

nels, productive research tasks, and recurring mechanisms—rather than identical outputs or replication in another market; the single-run configuration comparisons accordingly remain descriptive. Once a hypothesis is discovered, however, its symbolic expression and empirical protocol are fixed, making its reported results reproducible and allowing the hypothesis itself to be tested in future samples and other markets.

The remainder of the paper proceeds as follows. Section 2 situates the paper in the relevant literatures. Section 3 describes the symbolic language, evaluation pipeline, and discovery loop. Section 4 outlines the experimental design. Section 5 reports the discovery results. Section 6 subjects the discoveries to the rigorous testing framework. Section 7 analyzes the reasoning and revision traces through case studies. Section 8 discusses interpretation, limitations, and implications for AI-assisted research in finance.

2 Related Literature

2.1 Search Denominators, Data Snooping, and the Factor Zoo

Concern about selecting a successful return predictor from a larger set of unsuccessful attempts predates the modern factor zoo. Lo and MacKinlay (1990) show that asset-pricing tests can be severely distorted when properties of the same data are used to construct the test portfolios or statistics. Sullivan et al. (1999) evaluate technical trading rules against the full universe from which they were selected, demonstrating that an apparently successful rule can lose its significance once data snooping is taken into account. White (2000) formalizes this logic in the Reality Check, which tests whether the best model encountered during a specification search outperforms a benchmark after accounting for the other models examined. The common insight is that evidence about a selected winner cannot be separated from the search that produced it.

The factor-zoo literature brings this concern directly to the cross-section of expected returns. Harvey et al. (2016) show that the significance hurdle for a proposed factor depends on the number of factors tried and develop an approach for the empirically important case in which that number is unknown. Their analysis makes the hidden search denominator central: a conventional t-statistic may be persuasive when it comes from a single prespecified test but much less so when it is the best result among many attempts. Harvey (2017) connects the statistical problem to the organization of research. Unreported tests, incentives to obtain significant results, flexible specifications, and post hoc choices can all make the published record

more favorable than the underlying research process and thereby weaken reproducibility.

Subsequent work develops and compares methods for inference over large families of financial hypotheses. Harvey and Liu (2020) use a double-bootstrap procedure to calibrate both false and missed discoveries, recognizing that a procedure can avoid false positives only by sacrificing power if it is too conservative. Harvey et al. (2020) compare alternative multiple-testing procedures across financial applications and show that their relative performance depends on features such as power, dependence, and the composition of the candidate family. This literature does not imply that most published predictors must be false. Using 156 published long–short portfolios, Chen and Zimmermann (2020) estimate that publication-bias-adjusted returns are only modestly below in-sample returns and argue that the dispersion across predictors is too large to be explained by data-mined noise alone. Together, these studies establish both the need for search-aware inference and the importance of allowing for missed as well as false discoveries.

Existing procedures generally begin with a supplied family of candidate tests or, when the full family is unobserved, an assumed or statistically inferred number of tests, and ask how inference should be adjusted for multiplicity. The search process that generated this family—including unsuccessful hypotheses and the feedback that shaped later choices—typically remains hidden. Our design operates one step upstream by recording that process directly. Every executable hypothesis, successful or unsuccessful, enters the archive, making the realized search denominator observable rather than inferred from published winners. We then apply established multiple-testing methods to this observed population and examine how later hypotheses respond to earlier results. Our contribution is therefore not a substitute for multiple-testing inference, but an infrastructure that supplies its normally hidden input: the actual hypotheses examined and the adaptive path through which they were constructed.

2.2 Automated Discovery in Asset Pricing

Our study also relates to research that uses flexible statistical methods to organize large asset-pricing information sets. The machine-learning asset-pricing literature estimates return-prediction functions or stochastic discount factors from broad but supplied sets of characteristics and test assets (Gu et al., 2020; Freyberger et al., 2020; Chen et al., 2024; Kelly et al., 2025). This work shows how nonlinearities, interactions, and regularization can improve prediction and measurement in high-dimensional environments. Its principal object, however, is usually a predictive mapping over an observed characteristic universe, rather than an adaptive process

that proposes new economic hypotheses and revises their construction after receiving empirical feedback.

A closely related literature asks which factors or alphas carry incremental information once the large observed benchmark family is taken seriously. Feng et al. (2020) develop a model-selection method for testing whether a proposed factor adds explanatory power beyond a high-dimensional set of existing factors, while accounting for errors in selecting the benchmark model. Giglio et al. (2021) develop multiple-testing procedures for large alpha panels that remain valid with omitted factors and missing observations. Large-scale replication and factor-comparison projects likewise impose common construction and evaluation standards on published return predictors (Hou et al., 2020; Chen and Zimmermann, 2022). These papers provide natural tools for asking whether a candidate is distinct from known factors or whether any apparent alpha remains after many simultaneous tests. In each case, however, the empirical family is supplied to the procedure. Our setting adds the upstream question of how an AI system constructs that family through adaptive search.

Recent automated-discovery systems move closer to this upstream problem by generating mathematical expressions, trading rules, or candidate factors rather than estimating only a return-prediction function over fixed inputs (Zhang et al., 2020; Cui et al., 2021; Yu et al., 2023; Wang et al., 2025; Shi et al., 2025a; Tang et al., 2025; Shi et al., 2025b; Kou et al., 2025; Weng et al., 2026). These systems demonstrate that symbolic search, evolutionary methods, and LLM-based agents can scale alpha mining. Our paper complements that work by making the research process, rather than only the resulting winners, the unit of observation. Each hypothesis contains an economic mechanism, an executable accounting expression, and a sign predicted before evaluation; deterministic code then validates and grades it under fixed empirical rules. The archive retains unsuccessful hypotheses, critic assessments, revisions, and standardized outcomes, allowing us to study how evidence redirects later search.

2.3 AI and the Production of Research

A broader literature studies how computational systems can contribute to scientific discovery. Ludwig and Mullainathan (2024) treat machine learning as a tool for generating novel and interpretable hypotheses, emphasizing that hypothesis generation is a distinct research activity rather than an informal prelude to testing. Related work considers AI as a complement to scientific labor and as a system for generating or evaluating research ideas (Korinek, 2023;

Manning et al., 2024; Agrawal et al., 2026; Wang et al., 2023). Scientific applications such as AlphaFold and AI Feynman further illustrate that useful discovery can arise when flexible algorithms operate within domain-specific representations and evaluation rules (Jumper et al., 2021; Udrescu and Tegmark, 2020). Our laboratory follows the same organizing principle but studies a domain in which adaptive reuse of a single historical panel makes the statistical consequences of hypothesis generation especially important.

Within finance, LLMs have been used to extract return-predictive information from text, assist investors, and automate research tasks (Lopez-Lira and Tang, 2023; Dong et al., 2024; Bernard et al., 2026). These applications establish that language models can process financial narratives and perform parts of an analyst's workflow. Our object differs: the agent does not begin with an exogenously supplied text signal or a selected empirical result. It constructs executable accounting hypotheses, receives standardized evidence about each attempt, and uses that evidence in subsequent rounds. The resulting trace makes the production process itself empirically observable.

Novy-Marx and Velikov (2026) provide an especially important point of comparison. They show that LLMs can generate complete academic finance papers after empirically promising return predictors have been identified. Their pipeline mines more than 30,000 potential accounting signals, applies the Novy-Marx and Velikov (2024) *Assaying Anomalies* protocol to identify 95 signals that meet demanding empirical criteria, and then uses LLMs to produce multiple complete papers with distinct theoretical explanations and supporting citations. This contribution demonstrates the striking scale at which AI can produce finance research outputs. It also provides a cautionary demonstration of how alternative narratives can be written after the return evidence is known, thereby industrializing hypothesizing after results are known.

Our system moves upstream and intervenes earlier in the research process: agents state an economic mechanism and predicted sign before each hypothesis is tested, all successes and failures remain in the archive, and later revisions can be linked to the feedback that preceded them. This architecture makes the realized search denominator and the path of specification construction available for statistical scrutiny, economic interpretation, and replication.

3 Framework: AI-Guided Hypothesis Discovery

Large language models can generate plausible economic mechanisms at low marginal cost, but unconstrained search permits a model to vary an idea, its measurement, and its empirical test simultaneously. Our framework separates these choices. The AI research agents choose an economic mechanism, an executable expression, and a predicted return direction within a human-designed research environment; deterministic code enforces the admissible language, constructs the resulting signal, and applies the same empirical pipeline to every validated hypothesis. The agents therefore execute a bounded hypothesis-construction loop rather than the full empirical research process.

This division of labor assigns semantically rich tasks—mechanism formation, symbolic representation, directional prediction, and critique—to the agents while reserving data construction, estimation, and decision rules for fixed code. The design is related to evaluator–optimizer agent patterns (Schluntz and Zhang, 2024) and constrained program search in scientific discovery (Romera-Paredes et al., 2024). Its purpose is not simply to generate more characteristics, but to make model-generated hypotheses economically legible, empirically comparable, and traceable through post-search inference.

Figure 1 summarizes the framework. A human-designed environment fixes the hypothesis language, cross-theme research tasks, and empirical protocol. Within that environment, the proposer formulates hypotheses, the critic evaluates their economic logic and measurement before their current empirical outcomes are observed, and deterministic code validates and evaluates the resulting expressions. A task-specific archive carries standardized evidence from completed evaluations into the next generation. Once a run ends, post-search procedures operate outside the discovery loop and are indexed to the population for which each procedure is defined. Appendix A documents the implementation and measurement conventions.

3.1 Executable Economic Hypotheses

An agent proposal is the structured record produced before deterministic validation. We define the semantic core of the proposed hypothesis as

$$h_j = (m_j, e_j, d_j), \tag{1}$$

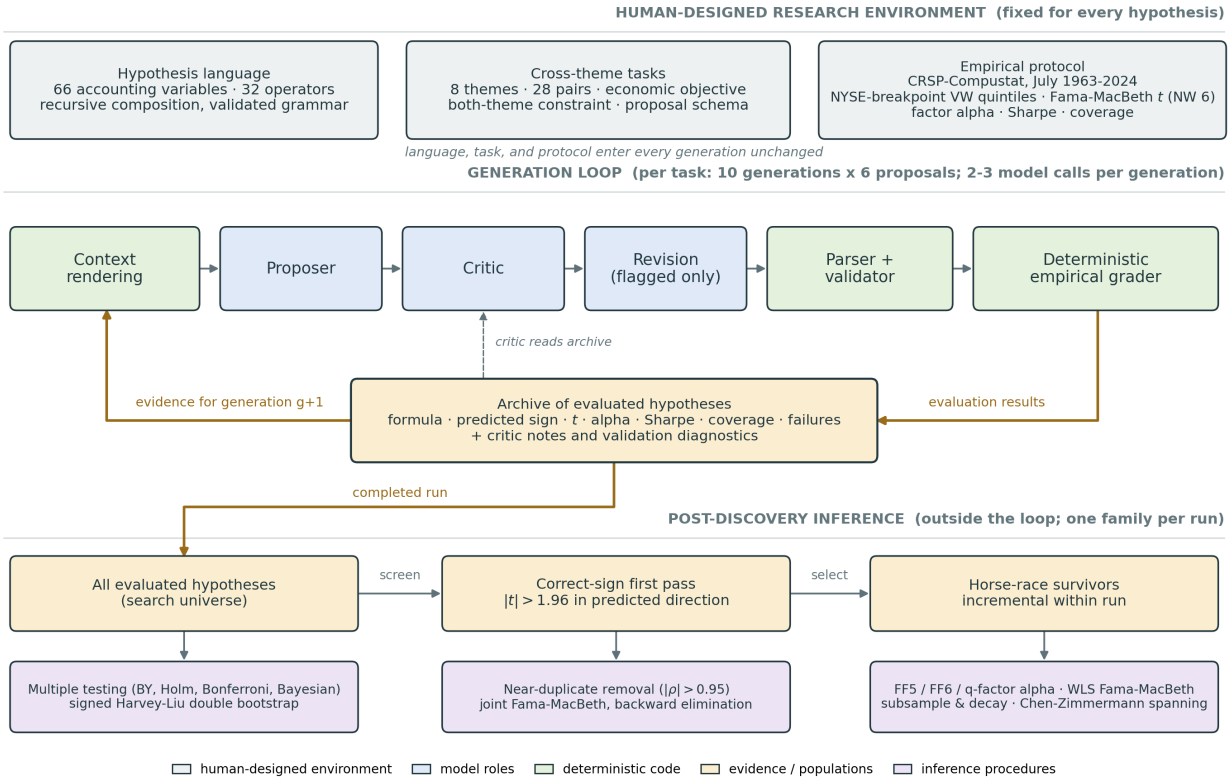


Figure 1: SignalLab discovery framework. The top panel defines the research environment held fixed across hypotheses. The middle panel shows the adaptive, task-specific generation loop: the proposer formulates hypotheses, the critic evaluates each current proposal before observing its current empirical outcome, deterministic code validates the expression, and the empirical grader returns standardized results to the archive. The bottom panel separates the run-level evaluated search universe, the correct-sign first pass, and the union of the 28 theme-pair horse-race survivor sets. Run-level multiple-testing procedures operate on the evaluated search universe, joint selection operates separately within each theme pair, and temporal, factor-model, and published-predictor tests operate on the horse-race survivors.

where m_j is the stated economic mechanism, e_j is an executable symbolic expression, and $d_j \in \{-1, +1\}$ is the return direction declared before hypothesis j is evaluated. Applying e_j to the firm-level inputs produces the signal $S_{it}^{(j)}$. The recorded hypothesis additionally contains a unique name, construction-type metadata, and a concise rationale, as detailed in Appendix A. Requiring the mechanism, expression, and direction serves three distinct purposes. The mechanism states the economic conjecture, the expression fixes exactly what is measured, and the declared direction prevents the agent from reversing the prediction after observing that hypothesis's empirical result.

Throughout the paper, *proposal* denotes the structured model output submitted before validation, and *hypothesis* denotes the mechanism–expression–direction triplet. We use *expression* or *formula* for e_j, *signal* for the realized firm-period values $S_{it}^{(j)}$, and *survivor* only with a named stage of the empirical funnel.

Expressions are written in a domain-specific language built from 66 annual accounting variables and 32 composable operators. The language admits ratios, differences, growth rates, temporal transformations, ranks, interactions, and industry-relative constructions, but not arbitrary code. For example,

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SUMRANK(IND_ADJ(RATIO(SUB(revt,cogs),at)),NEG(GROWTH(at,1)))        (2)
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combines industry-adjusted gross profitability with low asset growth. This is a cross-theme construction rather than an interaction in the narrow econometric sense. More generally, we use *cross-theme construction* to encompass additive composites, multiplicative interactions, conditional rules, and weakest-link functions. The language is flexible enough to express these economically motivated constructions but structured enough that the same evaluator can execute every expression without model-written data code. As in other scientific-search systems, the restricted representation removes malformed or incomparable outputs without prescribing which hypothesis should be found (Jumper et al., 2021; Romera-Paredes et al., 2024).

We organize the search over all $\binom{8}{2} = 28$ unordered pairs of eight themes: profitability, valuation, investment, accruals, quality, financing, intangibles, and distress. These themes span familiar dimensions of the cross section of expected returns. A deterministic validator requires each expression to contain at least one variable assigned uniquely to each theme in its pair. A variable shared across the two theme catalogs, including a generic denominator, cannot satisfy both requirements. Thus, a profitability–investment hypothesis must contain a theme-

specific profitability input and a theme-specific investment input. Here, "theme-specific" refers to membership in the prespecified variable catalogs; it is a syntactic validation rule, not a claim that an accounting quantity has only one economic interpretation. The restriction ensures that each validated expression combines information from both assigned themes. It does not itself establish novelty, which is assessed later using factor and published-predictor tests.

3.2 Agent Choices within a Fixed Research Environment

The framework separates choices delegated to the agents from rules fixed by the researchers before a run begins. The agents choose which mechanism to pursue, how to represent it using the available variables and operators, and which return direction to predict. Researchers fix the task definitions, admissible language, information timing, investable universe, missing-data conventions, estimators, and decision rules. During a run, the archive and the hypotheses generated in response to it evolve; the research environment and empirical grader do not.

This separation addresses a central comparability problem in open-ended model research. If an agent can choose a proposal-specific sample, alter missing-data treatment, or replace an estimator after observing results, measured performance confounds the economic content of the idea with changes in its test. In our framework, deterministic code applies common timing, sample, data-treatment, and estimation rules to every validated hypothesis. Expressions can still generate different coverage and empirical distributions, but agents cannot improve a proposal's measured performance by changing the evaluation protocol. The common pipeline therefore places hypotheses from different generations and model configurations on a common empirical scale.

The same boundary governs the configuration comparisons. A hypothesis proposed by one model provider has the same representation, validation rules, and empirical evaluation as a hypothesis proposed by another. Removing the critic changes the realized search process but leaves the task set, admissible symbolic language, validator, and grader unchanged. Because the alternative configurations are each executed once and the no-critique configuration also uses less model compute, the resulting comparisons are descriptive rather than causal estimates of model or critic effects. Section 4.5 describes these configurations in detail.

3.3 Critique before Empirical Evaluation

The primary search topology places a critic between proposal and empirical evaluation. The proposer first produces a batch of hypotheses using the assigned task and evidence accumulated in earlier generations. The critic observes the current proposals and the earlier archive, but not the current proposals' empirical outcomes. It assesses whether each expression implements its stated mechanism, whether the predicted direction follows from that mechanism, and whether the proposal materially differs from hypotheses already evaluated within the task. Flagged proposals return to the proposer for targeted revision. Unflagged proposals and revised versions of flagged proposals then pass through deterministic validation before reaching the empirical grader.

This timing separates economic critique from statistical selection. Because the critic sees earlier-generation evidence, it can identify repetition, a mismatch between a mechanism and its implementation, or a directional claim inconsistent with the stated logic. Because it does not see the current batch's outcomes, it cannot retain a current proposal merely because that proposal happens to produce a favorable statistic. Statistical selection begins only after the fixed grader evaluates a validated expression.

Within a given run, the archive is the state variable of the adaptive search. Let $\mathcal{J}_{k,g}$ denote the hypotheses from task k evaluated by the end of generation g, and let E_j denote the standardized aggregate empirical record returned for hypothesis j. The evaluated archive supplied to later generations of task k is

$$\mathcal{A}_{k,g} = \{(h_j, E_j) : j \in \mathcal{J}_{k,g}\}. \tag{3}$$

The broader process trail also retains proposer responses, critic notes, superseded proposals, parsing outcomes, and validation findings. Beginning with Generation 1, the proposer receives its task's evaluated archive together with recent critic and validation feedback; neither the proposer nor the critic receives firm-level observations. No archive or conversational state is shared across theme-pair tasks or runs. Later proposals can therefore condition on standardized results from earlier trials, but any response to those results must take the form of another explicit hypothesis and face the same validation and evaluation rules.

In operational terms, the archive supplies structured context for subsequent model calls (Anthropic, 2025). Its scientific role is to convert each task's sequence into a recorded experi-

mental history containing successful and unsuccessful evaluations. This record allows the trace analysis to link a later proposal to observable prior evidence and critique rather than treating the search as an unstructured conversation.

3.4 Adaptive Search and Inferential Populations

The generation loop is adaptive and in-sample. Later hypotheses respond to statistics computed from the same return history used to evaluate earlier hypotheses. A large discovery-stage statistic therefore describes what the search found within the historical panel; it is not, by itself, evidence from a prespecified test. At a minimum, post-search inference must account for the realized family of hypotheses produced by the search.

We distinguish three empirical populations. First, all empirically evaluated hypotheses with usable discovery statistics in a run form its realized testing family for run-level multiplicity analysis; the signed double bootstrap uses the corresponding executable hypotheses for which the monthly Fama–MacBeth slope series can be reconstructed. Second, hypotheses whose Fama–MacBeth statistic exceeds the discovery threshold in their declared direction form the correct-sign first pass, equivalently $d_j t_{\mathrm{FMB},j} > 1.96$. Within each run, these hypotheses compete separately within their assigned theme pair: signals with absolute average cross-sectional rank correlation above 0.95 are treated as near-duplicates, and the remaining candidates enter a joint Fama–MacBeth regression. The union of the 28 pair-specific horse-race survivor sets forms the run-level survivor population. Third, those horse-race survivors enter the temporal-stability, factor-model, and published-predictor tests. Multiple-testing adjustment and the horse race therefore address different comparison sets: the former uses the full run-level testing family, whereas the latter evaluates overlap among related constructions within an economic task.

These post-search procedures condition on the realized set of executable hypotheses. They do not rerun the full proposer–critic–revision architecture under the null and therefore should not be interpreted as end-to-end selective inference for the adaptive search algorithm. Observing the realized testing family makes a normally hidden input to multiple-testing procedures available; it does not eliminate the additional dependence created when later hypotheses respond to earlier outcomes.

The resulting funnel makes attrition, rather than the largest in-sample statistic, the organizing principle of the analysis. A first-pass hypothesis may be redundant within its theme pair, absorbed by known factors, concentrated in an early period, or spanned by published predic-

tors. Only a complete-protocol survivor has passed the full set of gates directed at these distinct threats to interpretation. Each run remains a separate testing family. The repeated primary runs use separately initialized model sessions and archives but the same historical return panel, so they are distinct search realizations rather than statistically independent samples. We use them to broaden the realized hypothesis space and assess convergence in search outcomes, while reporting their results separately rather than pooling hypotheses or correction thresholds.

4 Research Design and Empirical Protocol

Across all runs, we hold fixed the human-designed research environment: the variable and operator catalogs, expression grammar, theme-pair tasks, information-timing rules, investable-universe criteria, and empirical grader. The three primary runs repeat the same proposer–critic configuration from separately initialized sessions, whereas the comparison runs either remove the critique–revision subloop or replace the proposer model. This common environment places realized search outputs on a comparable empirical basis. Because each comparison configuration is executed only once and the no-critique configuration uses less model compute, however, differences across configurations are descriptive rather than causal estimates of model or critic effects.

4.1 Data, Sample, and Investable Universe

We combine monthly security data from the Center for Research in Security Prices (CRSP) with annual financial-statement data from Compustat using the CRSP–Compustat Merged (CCM) link table. The source histories begin in 1960, providing the observations required for lagged and rolling constructions before the return-prediction sample begins in July 1963. The sample ends in December 2024.

Our CRSP panel contains approximately 3.5 million security-month observations for roughly 26,000 ordinary common shares (share codes 10 and 11) listed on NYSE, AMEX, or NASDAQ. Returns incorporate available delisting returns and the fixed missing-delisting-return convention described in Appendix A.3 (Shumway, 1997). We define market equity as the absolute value of CRSP price multiplied by shares outstanding.

The Compustat panel uses annual industrial-format, standardized, domestic, consolidated records. We enforce CCM link dates, retain valid primary links, and resolve multiple records

mapping to the same security-year using the deterministic rules in Appendix A.3. The symbolic language draws on 66 variables from the balance sheet, income statement, statement of cash flows, and financing disclosures. Monthly Fama–French factors come from Kenneth French's data library (Fama and French, 1993, 2015). Post-search analyses also use the Hou–Xue–Zhang q-factor model (Hou et al., 2015) and the 209 firm-level predictors in the Chen–Zimmermann open-source library (Chen and Zimmermann, 2022).

The investable universe excludes financial firms and microcaps. We identify financial firms using CRSP SIC codes 6000–6999; their balance-sheet structure and the economic meanings of leverage, working capital, and investment differ materially from those of industrial firms. In each formation month, the microcap boundary is the 10th percentile of NYSE market equity, following the investability concerns emphasized by Hou et al. (2020). NYSE, AMEX, and NASDAQ securities above this threshold remain eligible, but NYSE securities alone determine portfolio breakpoints. Appendix A.3 reports the exact sample filters, link-resolution rules, and missing-data conventions.

4.2 Signal Measurement and Information Timing

Deterministic code evaluates each validated expression on the annual Compustat–CCM panel. The resulting firm-year values are winsorized within formation-year cross sections at the 1st and 99th percentiles. Accounting observations for fiscal years ending in calendar year y first become eligible in July of $y+1$ and remain in force through June of $y+2$. This schedule imposes a minimum six-month lag between fiscal year-end and the first portfolio-formation date.

We map each annual signal to a contiguous monthly panel beginning in July 1963. Within each month, nonmissing signal values are converted to percentile ranks centered at zero. Portfolio assignments and cross-sectional regressors are lagged one month relative to returns. Within an active month, a missing signal value receives the median rank of zero, as detailed in Appendix A.3. The design therefore relates a signal fixed in formation month t to returns in month $t+1$, and the agents cannot alter the information-timing or missing-data rules for an individual hypothesis.

Before investable-universe exclusions or cross-sectional filling, the pipeline rejects an expression whose nonmissing aligned coverage is below 1%. This permissive mechanical screen asks only whether the expression produces enough nonmissing aligned observations to enter empirical evaluation; it does not establish adequate coverage for the final investable sample or

for subsequent joint tests. Appendix A.3 specifies the coverage denominator, gap-aware annual lags, active-date boundaries, and the remaining alignment rules. The six-month eligibility convention governs the timing of the supplied Compustat observations; it does not by itself imply that the database is a vintage, as-originally-reported archive.

4.3 Discovery-Stage Estimands

We evaluate each resulting signal using two complementary views: a univariate monthly Fama–MacBeth regression and a value-weighted quintile portfolio sort. For hypothesis j, the primary discovery statistic is the slope from the monthly Fama–MacBeth regression (Fama and MacBeth, 1973),

$$R_{i,t+1} = a_{t+1} + b_{t+1}^{(j)} z_{i,t}^{(j)} + \varepsilon_{i,t+1}^{(j)}, \tag{4}$$

where $z_{i,t}^{(j)}$ is the month-t ranked signal. We report $\bar{b}^{(j)} = T^{-1} \sum_t b_t^{(j)}$ and its Newey–West t-statistic using six monthly lags (Newey and West, 1987; Petersen, 2008).

For the portfolio view, NYSE securities determine monthly quintile breakpoints, and all eligible securities are assigned using those breakpoints. Portfolio membership and market-equity weights are fixed in month t, and the value-weighted return in month $t+1$ is

$$R_{p,t+1} = \sum_{i \in p_t} \frac{\mathrm{ME}_{i,t}}{\sum_{\ell \in p_t} \mathrm{ME}_{\ell,t}} R_{i,t+1}. \tag{5}$$

The raw long–short return is Q5 minus Q1. When performance is evaluated relative to hypothesis j's declared direction d_j, we sign-orient the spread and its associated Sharpe ratio, alpha, and t-statistics so that a positive value denotes agreement with the prediction. Unless otherwise noted, reported portfolio-performance statistics are sign-oriented. The discovery-stage alpha uses the Fama–French five-factor model, and the Sharpe ratio is annualized.

We classify an empirically evaluated hypothesis as a first-pass success when its Fama–MacBeth statistic is finite, exceeds 1.96 in absolute value, and the estimated slope has the direction declared before evaluation; equivalently, $d_j t_{\mathrm{FMB},j} > 1.96$. The first-pass success rate is the number of such hypotheses divided by all hypotheses submitted to empirical evaluation in the relevant run, generation, or task, including submissions that end in execution errors. This rule is a discovery-stage screen rather than search-adjusted inference. First-pass hypotheses enter selection within their theme-pair task, while the full run-level family of hypotheses with usable statistics remains the relevant population for multiplicity analysis.

4.4 Theme-Pair Search

The search covers eight themes: profitability, valuation, investment, accruals, quality, financing, intangibles, and distress. These themes encompass prominent return-predictive mechanisms, including productive efficiency and investment (Novy-Marx, 2013; Fama and French, 2015; Hou et al., 2015), value and fundamental strength (Piotroski, 2000; Asness et al., 2019), accrual quality (Sloan, 1996; Richardson et al., 2005), external financing (Pontiff and Woodgate, 2008), and intangible capital (Peters and Taylor, 2017).

We instantiate all 28 unordered theme pairs. For each pair, a human-authored task prompt specifies an economic objective, theme-specific resources, and conceptual examples of possible cross-theme logic. Within that frame, the proposer chooses which mechanism to instantiate, how to encode it as an executable expression, and which return direction to predict. The prompts provide neither executable candidate formulas nor empirical results labeled as successful. As described in Section 3.1, a deterministic validator requires each expression to contain at least one variable assigned uniquely to each theme in its pair. Table B.2 reports the 28 task objectives.

Within a run, each theme pair is an operationally isolated search unit with its own proposer state, archive, and validation history; empirical feedback carries across generations within that pair but is not shared across pairs. This isolation does not imply statistical independence across the resulting signals. Each pair runs for 10 generations and requests six proposals per generation, corresponding to 60 requested slots per pair, 280 pair-generations, and 1,680 requested slots per run. A model response can contain more or fewer parseable final hypotheses than requested, and a parsed hypothesis may subsequently fail deterministic validation or empirical execution. We use *considered proposal* for the final parsed record submitted to validation, *validated hypothesis* for a record that passes the deterministic checks, and *empirically evaluated hypothesis* for a validated record submitted to the grader. An empirically evaluated hypothesis can end in an execution error rather than a usable statistic, so requested slots and these three realized counts can differ. Appendix A.2 defines the treatment of unparseable, invalid, or nonexecutable records.

4.5 Agent Configurations

Table 1 reports the six search runs. The primary configuration assigns `claude-opus-5` to both the proposer and critic and uses the propose–critique–revise topology described in

Section 3.3. We execute this configuration three times from fresh provider sessions and empty pair-level archives, denoting the resulting runs B1–B3. No archive or conversational state is shared across runs, and the resulting testing families are not pooled for inference.

The comparison runs vary the agent setup while preserving the research environment. Run A1 retains the primary proposer but removes both critique and revision. It therefore uses fewer model calls and substantially fewer output tokens, so the comparison bundles search topology with compute. Runs A2 and A3 replace the proposer with `gpt-5.5` and `gpt-5.6-sol`, respectively, while retaining `claude-opus-5` as the critic and preserving the propose–critique–revise topology. Across all six runs, the task set, number of generations, requested batch size, symbolic language, validator, data, sample filters, and empirical grader remain unchanged.

We use neither fine-tuning nor retrieval augmentation. Prompts contain the conceptual anchors described in Section 4.4, but no executable formulas, empirical performance information, or examples labeled as successful signals. All roles use the high-reasoning setting available from their respective providers. The six runs contain 10,080 requested proposal slots in total. Appendix A.2 documents provider sessions, context construction, model-call accounting, and the distinctions among requested, considered, validated, and evaluated proposals.

4.6 Post-Search Selection and Inference

Post-search procedures follow the inferential populations defined in Section 3.4, and each run remains a separate testing family. Run-level multiplicity analyses use all empirically evaluated hypotheses with usable Fama–MacBeth statistics. We report Bonferroni and Holm family-wise-error adjustments (Holm, 1979), the Benjamini–Yekutieli (BY) false-discovery-rate procedure (Benjamini and Yekutieli, 2001), and posterior null probabilities under the moderate prior described in Section 6 and Appendix A.6, following Harvey (2017). Bonferroni and Holm are reported as diagnostics; Section 6 identifies the criteria that enter the cumulative survival protocol. A signed Harvey–Liu double bootstrap estimates false-discovery behavior from the monthly slope matrix of the executable search population (Harvey et al., 2016; Harvey and Liu, 2020). These procedures adjust the recorded run-level family rather than only the hypotheses that clear the initial screen.

Separately, first-pass hypotheses compete within each run and theme pair. We first remove near-duplicates using average monthly cross-sectional rank correlation, retaining the hypothesis with the larger sign-oriented univariate statistic when absolute correlation exceeds 0.95. We

then estimate a joint Fama–MacBeth regression and apply backward elimination until each retained hypothesis has an estimable conditional t-statistic above 1.96 in its declared direction. This horse race identifies incremental conditional return association relative to other constructions generated for the same economic task; it does not replace run-level multiplicity adjustment or establish statistical independence or economic novelty.

The cumulative protocol intersects the horse-race survivors with the BY false-discovery-rate, posterior-null-probability, and signed-double-bootstrap requirements; temporal-stability and non-decay requirements; direction-consistent alphas under the Fama–French five-factor model, the six-factor model that augments it with momentum, and the Hou–Xue–Zhang q-factor model; and conditional tests against selected predictors from the Chen–Zimmermann library. The temporal analysis uses the full sample, 1963–1990, 1991–2024, periods before July 2000 and after June 2000, the post-2010 period, and non-recession months, together with rolling 60-month regressions used to classify decay. These fixed calendar splits are stability diagnostics rather than signal-specific pre- and post-publication periods or prospective out-of-sample tests. Market-equity-weighted Fama–MacBeth estimates are reported as diagnostics rather than survival gates.

For each candidate reaching the published-predictor stage, the spanning procedure selects up to ten predictors with the largest absolute average cross-sectional rank correlations, subject to the joint-coverage, rank, and conditioning requirements in Appendix A.6. The candidate must retain its declared direction and clear the specified conditional-statistic threshold after the selected controls enter. Appendix A.6 provides the exact thresholds, decay rule, and control-selection algorithm.

5 Discovery-Stage Evidence

This section describes the outputs of the search before applying the cumulative post-search tests. We first report the discovery funnel and within-run adaptation, then examine heterogeneity across economic tasks, descriptive differences across agent configurations, and cross-run repeatability. The three primary runs provide the main repeated-run evidence; the comparison runs are treated separately. Section 6 applies run-level multiple-testing procedures to each realized testing family and evaluates the horse-race survivors using temporal-stability, factor-spanning, and published-predictor tests.

5.1 Discovery Funnel and First-Pass Yield

The primary runs produce similar aggregate discovery funnels. Of 1,680 requested proposal slots per run, 1,603–1,607 hypotheses reach empirical evaluation after parsing and deterministic validation (Table 2). Between 949 and 1,001 satisfy the prespecified sign and Fama–MacBeth threshold. The corresponding first-pass success rate—computed over all hypotheses submitted to empirical evaluation, including execution errors—ranges from 59.1% to 62.3%, with a cross-run median of 60.0%.

The within-task horse races produce much sharper attrition. The 949–1,001 first-pass hypotheses yield 73, 78, and 82 survivors in B1, B2, and B3, respectively. These survivors retain conditional return association relative to other hypotheses generated for the same theme-pair task. Thus, high first-pass productivity does not imply that an equally large set retains incremental conditional content.

Deterministic validation removes 4.3–4.6% of parsed proposals. Most rejections violate the cross-theme requirement and occur in Generation 0, before the archive contains empirical feedback; such failures become uncommon in later generations. All 28 theme-pair tasks nevertheless produce at least one horse-race survivor in each primary run. The repeated runs therefore exhibit similar aggregate yield, even though the specific hypotheses underlying that yield can differ substantially.

5.2 Within-Run Adaptation Across Generations

Figure 2 shows how discovery-stage outcomes change as later generations observe a larger archive of prior results. Because Generation 0 contains fewer evaluated hypotheses and is disproportionately filtered by the cross-theme validator, we center the comparison on Generations 1–9. Across the three primary runs, the median first-pass success rate rises from 43.4% in Generation 1 to 74.3% in Generation 9. All three runs display the same broad upward pattern after the initial archive is formed.

The strongest observed statistic does not weaken as the first-pass rate rises. The median of the largest sign-oriented t_{FMB} in each generation increases from 5.5 in Generation 0 to 7.2–7.7 in Generations 6–9. Because this measure is a within-generation maximum and depends partly on the number of evaluated hypotheses, we interpret it descriptively. At minimum, the higher later-generation success rates are not accompanied by lower maximum statistics.

The comparison runs reinforce the distinction between within-sample adaptation and the

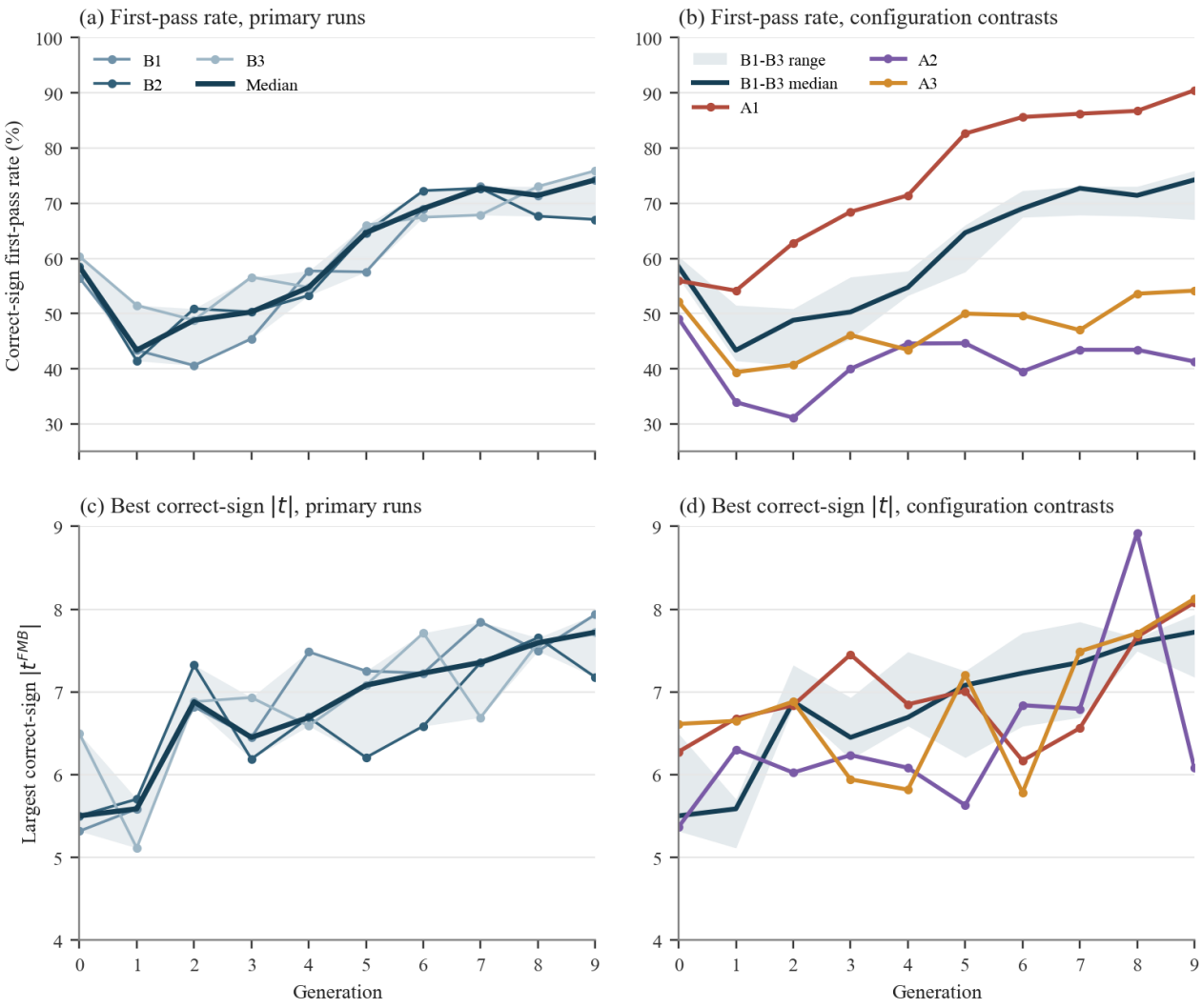


Figure 2: Discovery-stage outcomes by generation. Panels (a) and (c) report the three primary runs and their median; shaded regions show the full B1–B3 range. Panels (b) and (d) compare the three single-run alternative configurations with the primary-run median and range. The upper panels report first-pass success rates, and the lower panels report the largest sign-oriented t_{FMB} in each generation.

role of critique. The no-critique run dips slightly after Generation 0 and then rises steadily to a 90.5% first-pass rate in Generation 9; its maximum statistic fluctuates but ends higher at 8.1. The alternative-proposer runs end at lower first-pass rates of 41.3% and 54.2%, although their maximum statistics approach the primary-run range. These paths are consistent with the archive playing an important role in how proposals adapt to prior empirical outcomes. They do not isolate causal effects of the archive, proposer, or critic, because each comparison configuration is executed once and differs in realized model compute.

5.3 Heterogeneity Across Economic Tasks

Discovery-stage productivity varies substantially across the 28 theme-pair tasks. Figure 3 reports the primary-run median and range of the first-pass success rate and number of horse-race survivors for each task; Appendix Table B.3 provides the run-level results.

Valuation×Intangibles is the most consistently productive task, with a median first-pass success rate of 90.0%. Profitability×Accruals and Profitability×Quality also have median rates near 80%. By contrast, Financing×Distress and Valuation×Distress have median rates of 31.0% and 36.8%. This contrast persists when results are aggregated by theme within the task design: profitability tasks have a median rate of 69.8%, compared with 40.4% for distress tasks (Appendix Table B.4). Horse-race survivor counts vary much less across themes, indicating that the principal difference lies in how frequently hypotheses clear the univariate screen rather than in how many retain within-task conditional content.

The task ranking is moderately repeatable: pair-level first-pass ranks have cross-run Spearman correlations of 0.486–0.535. First-pass success and conditional yield nevertheless capture different margins. Financing×Intangibles, for example, has a median first-pass rate of 48.3% but four median horse-race survivors, whereas Profitability×Quality has an 80.0% first-pass rate but only one. Task productivity therefore depends on whether it is measured at the univariate or conditional stage.

5.4 Descriptive Configuration Comparisons

Figure 4 contrasts first-pass productivity with within-task incremental-survivor yield across agent configurations. These comparisons are descriptive. A1–A3 each contain one run, and A1 removes both critique and revision while also using fewer model calls. The estimates therefore

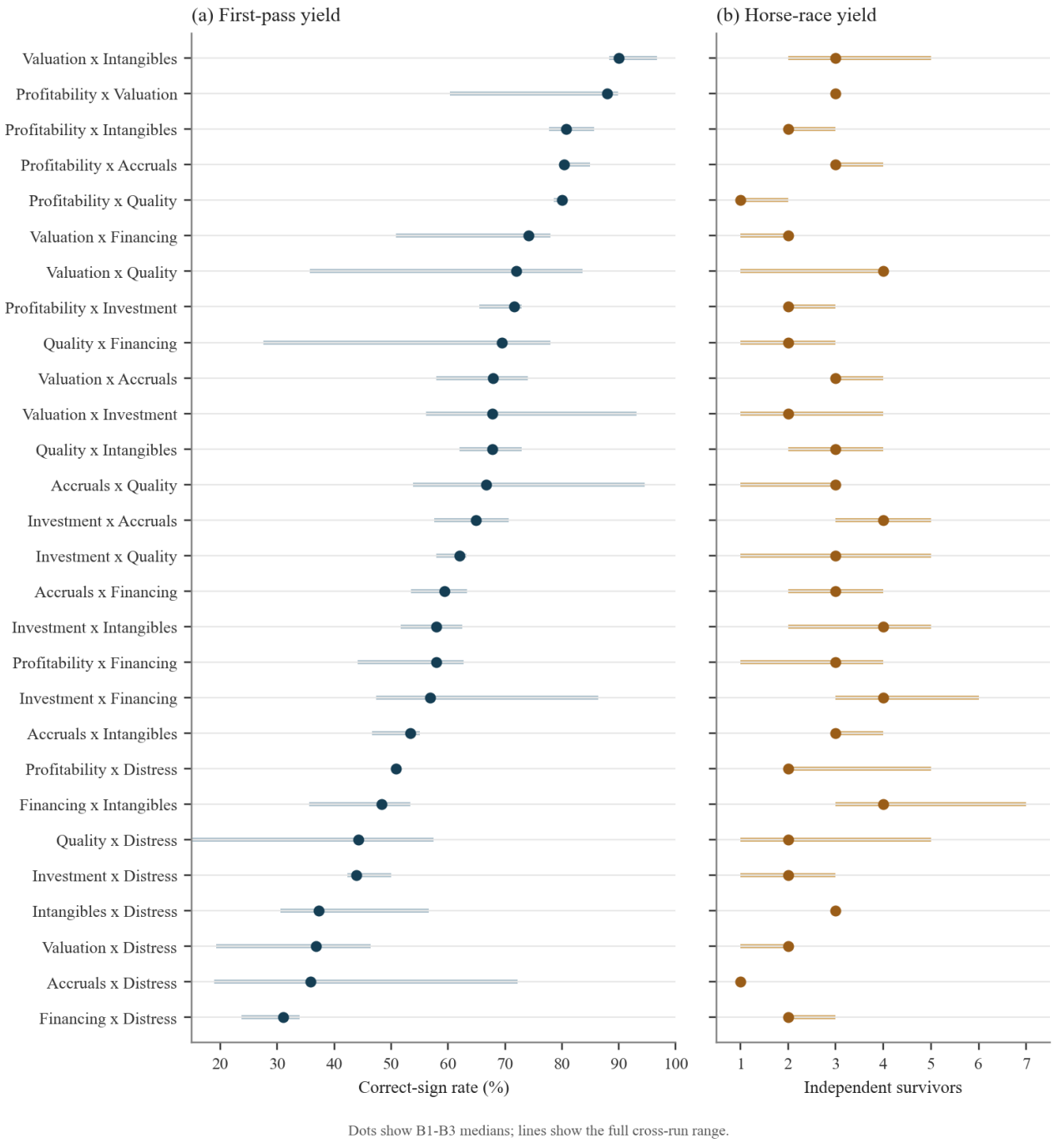


Figure 3: Discovery-stage outcomes across the 28 theme-pair tasks. Panel (a) reports first-pass success rates, and Panel (b) reports horse-race survivors. Dots denote B1–B3 medians, horizontal lines show the full range, and tasks are ordered by their median first-pass rate.

characterize realized search outputs under a common task and evaluation environment; they do not identify causal effects of critique or model identity.

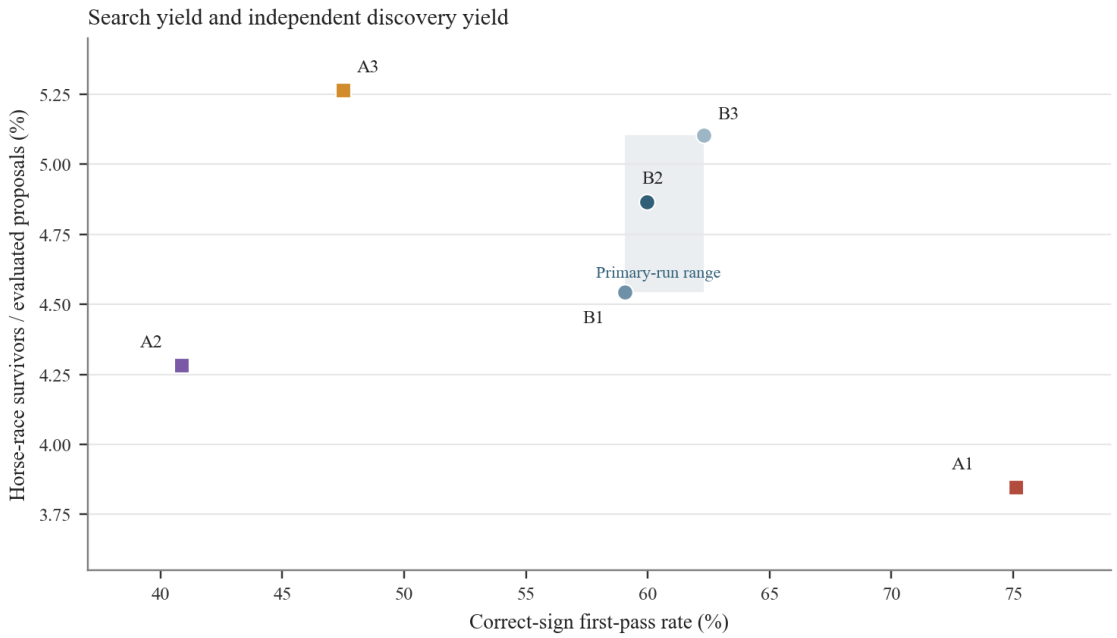


Figure 4: First-pass yield versus within-task incremental-survivor yield. The horizontal axis reports the first-pass success rate, and the vertical axis reports horse-race survivors divided by evaluated hypotheses. Circles denote the three primary runs, squares denote the single-run comparison configurations, and the shaded rectangle shows the B1–B3 range on both dimensions.

The no-critique run, A1, has the highest first-pass rate, 75.1%, and produces 1,211 first-pass hypotheses. Yet only 62 survive its within-task horse races, equal to 3.85% of evaluated hypotheses and below the primary-run range of 4.54–5.10%. Its additional first-pass hypotheses therefore do not translate into additional within-task survivors. This outcome is consistent with greater concentration in variants of previously successful constructions, but the single-run comparison does not establish critique as the cause.

An outcome-blind semantic diagnostic supports this interpretation. A1 uses an average of 6.82 coded mechanisms per task, compared with 7.57–7.75 in the primary runs, and its within-task Herfindahl index is 0.334, compared with 0.210–0.236. Because a higher Herfindahl index denotes greater concentration in fewer mechanisms, A1 covers fewer mechanisms and concentrates more heavily within them (Table 5). Across B1–B3, the critic flags 49–53% of requested proposal slots and triggers a revision in 839 of 840 pair-generations. These observations show that critique is used frequently and is associated with broader realized mechanism coverage; they do not quantify its causal contribution.

The alternative proposers illustrate the converse pattern. A3 records a first-pass rate of 47.5% but produces 85 horse-race survivors, above the primary-run range; A2 produces 69 survivors from a 40.9% first-pass rate. Thus, the number of hypotheses crossing a univariate threshold is not sufficient to characterize search output: a lower first-pass rate can coexist with a broader set of hypotheses that retain conditional content within their tasks.

5.5 Within-Task Survivors and Cross-Run Repeatability

Table 6 illustrates three high-Sharpe horse-race survivors from each primary run, selected only when the five-factor alpha has the declared direction. Appendix Table B.6 reports their expressions. Because the table selects realized winners after adaptive search and within-task selection, it is descriptive rather than inferential.

Cross-run repeatability depends on the level of comparison. Exact recurrence is rare: only one to three normalized expressions reappear in any pair of primary-run evaluation sets, and no horse-race survivor recurs verbatim. Recurrence is more common at the level of accounting inputs: 128–149 variable sets are shared across each pair of runs, including four or five sets represented among their survivors (Table 3). The runs therefore return to some of the same ingredients while combining them through different operators and functional forms.

To examine broader economic recurrence, we apply outcome-blind mechanism coding to all 9,656 evaluated hypotheses across the six runs and restrict the recurrence analysis to B1–B3. Two independent coding passes agree on 88.8% of records ($\kappa = 0.887$); Appendix A.8 documents the codebook and adjudication procedure. At the task–mechanism–sign level, 198 labels appear in all three primary runs, 90 in two runs, and 88 in only one. Among horse-race survivors, however, only eight labels recur in all three runs, while 38 appear in two and 94 in one (Table 4).

These comparisons reveal broad economic convergence but not exact reproducibility. The task–mechanism–sign taxonomy is deliberately coarse and finite, and each task generates many hypotheses; overlap at this level therefore shows that independent runs revisit similar economic claims, not that they recover the same formulas or survivors. The discovery funnels and task rankings are comparatively stable, whereas the identities of the surviving hypotheses remain strongly path dependent. Section 6 next asks which realized survivors withstand the cumulative post-search tests.

6 Cumulative Post-Search Validation

Section 5 reports 449 run-specific horse-race survivor records across the six search runs. We next intersect those records with a cumulative post-search protocol. Within each run, multiplicity statistics are computed from the full family of hypotheses with usable Fama–MacBeth statistics, and the signed double bootstrap uses the hypotheses for which monthly slopes can be reconstructed; the resulting decisions are then mapped to the horse-race set. Of the 9,656 empirically evaluated hypotheses, 9,645 yield usable full-sample statistics for the multiplicity analysis.[2] Eleven records clear the complete protocol: six from the three primary runs and five from the comparison runs.

The protocol addresses multiplicity within the recorded testing family, temporal instability, exposure to standard factors, and incremental information relative to selected published predictors. Our post-search procedures condition on the realized set of executable hypotheses. They do not rerun the full proposer–critic–revision architecture under the null and therefore should not be interpreted as end-to-end selective inference for the search algorithm. The temporal tests likewise use data observed by the discovery loop and are robustness diagnostics rather than prospective out-of-sample tests.

6.1 A Cumulative Evidentiary Standard

Following within-task horse-race selection, a record must clear seven cumulative gates: (1) the run-level Benjamini–Yekutieli (BY) false-discovery-rate procedure at $q = 0.05$; (2) the posterior-null-probability criterion; (3) the signed double-bootstrap threshold; (4) significance in the declared direction in at least four of seven sample windows; (5) the non-decay requirement; (6) significant, direction-consistent alphas under all three factor models; and (7) the published-predictor conditional test. Failure at one gate cannot be offset by strength at another.

The first three decisions are made at the run-family level before they are intersected with the 449 horse-race records. A record must survive the BY procedure (Benjamini and Yekutieli, 2001), have a posterior null probability below 5% under the moderate prior defined in Appendix A.6 (Harvey, 2017), and exceed the threshold from the signed double bootstrap of Harvey and Liu (2020). The bootstrap is calibrated separately for each run to target an estimated false-discovery rate of 5%; all six calibrations yield a sign-oriented t-statistic threshold

[2]The run-level usable counts are 1,603, 1,599, 1,607, 1,610, 1,612, and 1,614 for B1, B2, B3, A1, A2, and A3, respectively. Each run remains a separate testing family; totals across runs are descriptive.

of 2.50. Appendix A.6 reports the input p-value convention, the minimum-Bayes-factor construction and its prior odds, and the bootstrap calibration. BY is designed to control the false discovery rate under arbitrary dependence when the individual p-values are valid; in this adaptive setting, it provides multiplicity-adjusted evidence within the recorded testing family rather than guaranteed control for the search algorithm. Bonferroni and Holm adjustments are retained as family-wise-error diagnostics but do not enter the cumulative classification.

The temporal gates use seven fixed windows: the full sample, 1963–1990, 1991–2024, the periods before July 2000 and after June 2000, January 2011–December 2024, and non-recession months. A record must satisfy the declared-direction threshold in at least four windows and must not be classified as declining under the rolling 60-month procedure detailed in Appendix A.6. Because the windows overlap and differ substantially in length, the four-of-seven rule is a robustness screen rather than a collection of independent tests; it excludes candidates that fail in most windows but does not establish out-of-sample stability. The rolling analysis is similarly a diagnostic for declining historical performance, motivated by evidence that published return predictors attenuate over time (McLean and Pontiff, 2016).

The factor gate requires $d_j t_{\alpha,j} > 1.96$ under each of the Fama–French five-factor model, the six-factor model that augments it with momentum, and the Hou–Xue–Zhang q-factor model (Fama and French, 2015; Hou et al., 2015). Finally, the record must retain its declared direction and a conditional statistic above 1.96 after controlling for up to ten selected nearby predictors from the Chen–Zimmermann library (Chen and Zimmermann, 2022). The controls are chosen using cross-sectional rank correlations subject to the coverage and conditioning requirements in Appendix A.6. A record cannot pass if a required factor-model or published-predictor regression is not estimable. Market-equity-weighted Fama–MacBeth estimates remain sensitivity diagnostics rather than survival requirements.

6.2 Where Attrition Occurs

Table 7 reports the principal result:

$$449 \longrightarrow 325 \longrightarrow 207 \longrightarrow 103 \longrightarrow 11.$$

The first transition intersects the 449 horse-race records with the run-level BY decisions computed from the full usable families, leaving 325. Within this intersection, the Bayesian, signed

double-bootstrap, and four-of-seven requirements are nonbinding: all 325 clear them. This equality is a feature of the selected population, not evidence that the procedures are equivalent or that the Bayesian and bootstrap screens do not bind elsewhere in the full testing families.

Temporal instability provides the first substantial separation. The post-2010 window is the most selective, retaining 115 of the 325 records. The rolling-window procedure classifies 201 records as stable, six as strengthening, and 118 as declining; excluding the last group reduces the population to 207. Because the agents received feedback from the full historical panel, these counts describe stability within the discovery sample rather than performance on newly realized returns.

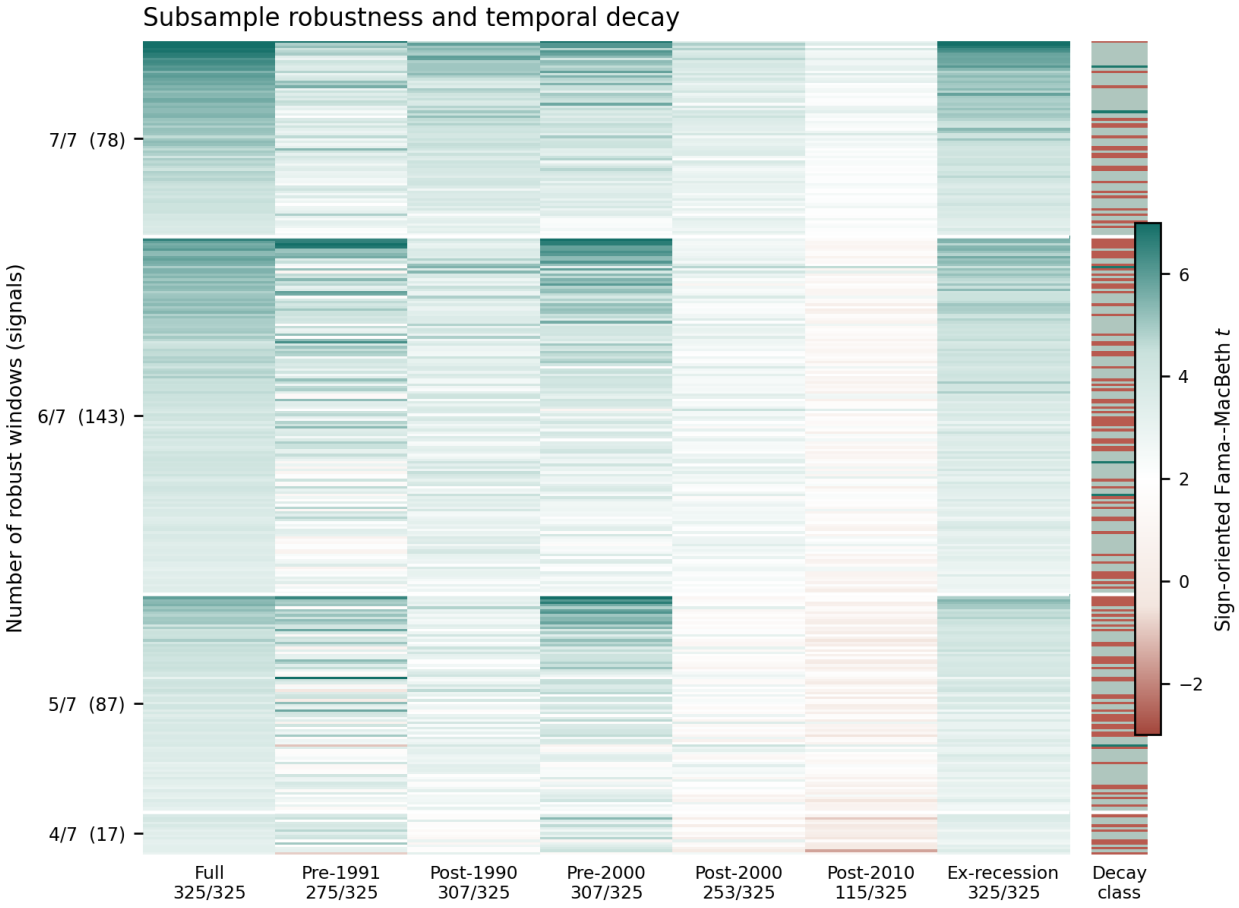


Figure 5: Temporal robustness of the 325 horse-race records that also clear the run-level multiplicity gates. Cells report sign-oriented Fama–MacBeth statistics across the seven fixed windows, and the side strip reports the rolling-window classification. These windows are robustness diagnostics rather than prospective out-of-sample tests.

Of the 207 non-declining records, 115 clear the Fama–French five-factor threshold, 111 clear

the six-factor threshold, and 110 clear the q-factor threshold when the models are considered separately; 103 clear all three. One record does not yield an estimable alpha suite and is counted as a failure. Thus, approximately half of the temporally admissible records have long–short returns that overlap sufficiently with standard factor exposures to fail at least one benchmark. These tests establish that overlap; they do not identify its economic source.

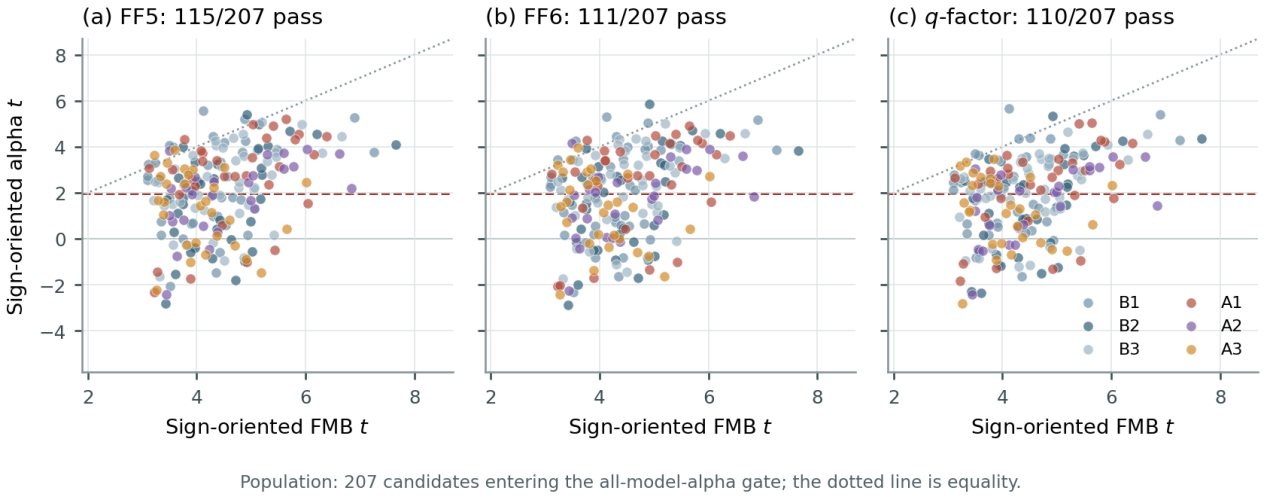


Figure 6: Factor-model tests for the 207 non-declining records. Each panel compares the sign-oriented full-sample Fama–MacBeth statistic with the corresponding factor-alpha statistic. The horizontal dashed line marks $t_\alpha = 1.96$, and the dotted line is the 45-degree line. The protocol requires a record to clear the threshold under all three models; 103 do so.

Conditional tests against selected published predictors impose the sharpest screen. Of the 103 records entering this stage, 101 yield estimable regressions, 11 retain a direction-consistent conditional statistic above 1.96, and two fail because no valid conditional regression can be estimated. For every estimable record, the absolute conditional statistic is below the corresponding absolute univariate statistic on the common sample; only three remain above 3.00. Many records that survive multiplicity and factor adjustment therefore contain information already represented by their selected published comparators.

Terminal counts should not be used to rank the agent configurations. Each primary run contributes two complete-protocol survivors, whereas the no-critique, GPT-5.5-proposer, and GPT-5.6-sol-proposer comparisons contribute four, one, and zero, respectively. These are small counts, and each comparison configuration is observed only once. The architecture evidence in Section 5.4 therefore remains descriptive and is based on the broader discovery populations.

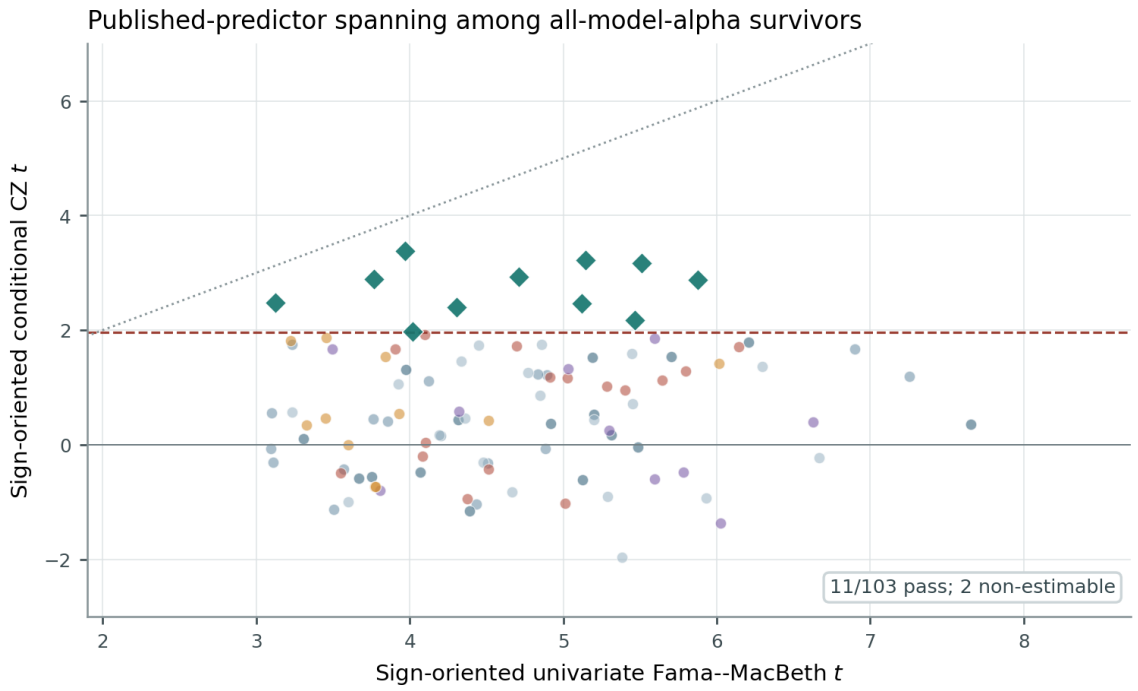


Figure 7: Conditional tests against selected published predictors. The figure compares each record's sign-oriented univariate statistic on the common estimation sample with its conditional statistic after up to ten nearby Chen–Zimmermann predictors enter. The horizontal dashed line marks 1.96, the dotted line is the 45-degree line, and diamonds identify the 11 complete-protocol survivors. Two additional records are non-estimable because of insufficient joint coverage.

6.3 Complete-Protocol Survivors

Table 8 reports the 11 complete-protocol survivors, and Appendix Table B.9 provides their executable expressions. Their full-sample Fama–MacBeth statistics range from 3.12 to 5.88. Each is significant in at least five of the seven fixed windows, and five are significant in all seven. The smallest absolute alpha statistic across the three factor models ranges from 2.25 to 4.35, while the conditional statistics after the selected Chen–Zimmermann predictors enter range from 1.98 to 3.38.

Coverage in the final conditional tests is uneven. Four of the 11 survivors have Chen–Zimmermann estimates based on only 24–28 monthly cross sections because joint coverage with the selected controls sharply shortens the common sample; the other seven use 437–737 months. We therefore interpret the short-sample conditional statistics as evidence within their available common samples, not as full-period spanning or conclusive evidence of novelty. Table 8 reports the valid-month count for each record. As a separate sensitivity diagnostic, seven of the 11 survivors retain a direction-consistent market-equity-weighted Fama–MacBeth statistic above 1.96; this diagnostic does not alter the declared complete-protocol set.

The survivor set requires a disciplined interpretation. First, the protocol adjusts for multiplicity within each recorded testing family and imposes several demanding conditional tests, but it does not transform an adaptively selected hypothesis into a prespecified experiment. A genuine out-of-sample test would freeze the models, prompts, expressions, data conventions, and empirical protocol before new returns are realized and would evaluate those returns prospectively without further revision. Second, survival against selected published predictors establishes incremental conditional content relative to those controls on the common estimation sample, not that every underlying variable or economic idea is new. Third, the 11 rows are run-specific records rather than necessarily distinct anomaly families; Signals 1 and 2, for example, are alternative expressions of the same build-versus-buy mechanism.

Within the historical panel, 11 run-specific records clear every declared gate. They remain candidates for prospective validation and external replication. Section 7 next uses the recorded discovery traces to examine how the agents developed several of these hypotheses.

7 Learning from the Recorded Discovery Process

Section 6 evaluates the outputs of the search. This section instead uses recorded proposals, critic assessments, revisions, and empirical outcomes to examine how several surviving hypotheses took shape. We ask what changed in the construct being measured, the conditions attached to its economic interpretation, or the functional role assigned to its components. The evidence concerns observable changes in the research record, not the model's latent cognitive process.

We examine four trajectories selected from the 11 complete-protocol survivors because the relevant transitions are especially clear. Signals 1 and 2 form one build-versus-buy lineage; Signal 4 conditions an ambiguous investment pattern on operating performance; Signal 6 combines familiar characteristics through a weakest-link rule; and Signal 10 changes the role assigned to an accrual measure after a mixed empirical result. The first three trajectories contain revisions following critic feedback, whereas Signal 10 comes from the no-critique run and provides an archive-only sequence without critic intervention. These cases are selected ex post and illustrate possible search operations rather than their frequency, average value, or the causal effect of critique.

7.1 Four Changes in Hypothesis Construction

Construct repair and mechanism extension: Signals 1 and 2. The initial build-versus-buy proposal combined persistent cash profitability with five-year acquisition spending divided by the larger of five-year R&D spending and depreciation. The proposal described the ratio as purchased relative to internally developed growth, but the critic noted that depreciation usually determined the denominator. The expression therefore measured acquisitions largely relative to capital consumption rather than the stated growth-sourcing construct.

The revision replaced the denominator with acquisition spending plus R&D spending, producing an acquisition share of those two measured channels, and retained persistent cash profitability. The economic idea remained intact, but the expression changed because the initial denominator computed the wrong object. In the next generation, Signal 2 represented the same mechanism through a different financial-statement contrast: five-year cumulative R&D spending, which is largely expensed as incurred, versus intangible assets recognized on the balance sheet. Neither quantity is a pure measure of internally developed or acquired capability, and the expression does not estimate a stock of internally generated knowledge capital.

It instead exploits their different reporting footprints. The sequence therefore contains both a construct repair and a representational extension, and Signals 1 and 2 are best viewed as related implementations of one mechanism rather than independent economic discoveries.

Conditional reformulation of an ambiguous investment pattern: Signal 4. A relative increase in R&D versus capital expenditure can reflect a deliberate shift toward knowledge-based investment, but it can also arise when a constrained or weakening firm reduces physical investment. The critic flagged variants that changed windows or appended mechanical gates without resolving this ambiguity. The revised expression combines the relative change in R&D versus capital expenditure with gross-profit growth. Gross-profit growth does not establish successful innovation; it supplies a separate operating-performance condition. The revision thus changes the question from whether an R&D tilt predicts returns to when that accounting pattern is informative, producing a sharper conditional hypothesis rather than another variation in timing or scaling.

Corroboration through a weakest-link rule: Signal 6. In another trajectory, the critic flagged a proposal as too close to familiar profitability constructions. The revision introduced an R&D-expensing channel and changed the logical structure of the hypothesis. Its first component adds current R&D expense to operating income before scaling by assets. We call this R&D-expense-adjusted profitability, not economic profitability, because the expression neither constructs an R&D capital stock nor specifies an amortization schedule. Its second component measures noncash current assets per dollar of sales, with a lower value serving as a separate balance-sheet condition.

Signal 6 ranks both components within industry and applies `MIN`, so the weaker component determines the score. High R&D-expense-adjusted profitability cannot offset high noncash current-asset intensity, and balance-sheet leanness cannot offset weak profitability. The research move lies in making familiar characteristics jointly necessary rather than additive. Consistent with the familiarity of its first component, the expression has a rank correlation of 0.55 with its closest published comparator, an R&D-adjusted operating-profitability predictor, yet the complete expression retains conditional predictive content and survives the protocol. This result does not isolate which component or operator produces that content; it shows that the final conjunction is not absorbed by the selected controls.

Reassigning a variable's functional role: Signal 10. Signal 10 comes from the no-critique configuration, so the relevant feedback is the empirical archive available to the next generation. An earlier accrual-gap hypothesis produced an almost flat Fama–MacBeth slope ($t = 0.20$) but a large long–short alpha in its declared negative direction (-7.0%, $t_\alpha = -4.63$). The next proposal's recorded rationale cites this disparity and describes the accrual measure as informative in the tails but not monotone across the full cross section. The disparity could also reflect differences in weighting, outliers, or the two estimators, so the record documents the proposal's response rather than validating that diagnosis.

The subsequent expression combines the accrual measure with trough cash-flow coverage and net cash, applying `MIN` so that the weakest dimension determines the score. The accrual measure no longer serves as a stand-alone continuous predictor; it becomes one binding condition in a broader rollover-resilience hypothesis. This trajectory shows that the archive can be followed by a change in what a familiar variable is asked to do, even without a critic intervention.

All four trajectories culminate in complete-protocol survivors reported in Table 8; their exact expressions and full validation evidence appear in the appendix and Section 6.

7.2 Aggregate Patterns in Critic Feedback

The cases identify consequential changes but do not show what the critic reviews more generally. We therefore classify all 8,398 structured critic judgments from the five proposer–critic runs (B1–B3 and A2–A3; A1 has no critic) using a frozen taxonomy. Each judgment receives one mutually exclusive dominant category and up to two nonexclusive secondary categories. The coding sessions do not observe the target proposal's empirical outcome or any post-search result. Two independent GPT-5.6 Terra sessions code the judgments, and a third session adjudicates disagreements; Appendix A.9 reports the definitions and reliability statistics.

Figure 8 and Table 9 report the distribution. Of the 8,398 judgments, 8,101 contain a material objection or caveat. Novelty or redundancy is the dominant concern in 42.9%. Measurement, functional form, and attribution of the proposed interaction together account for another 39.3%, while data feasibility and failure to respond to an archive diagnostic account for 14.6%. The critic therefore does more than screen for familiar formulas: it frequently questions whether an expression measures, distinguishes, and updates its stated mechanism. These counts describe concerns raised by the critic, not independently verified defects or evidence that critique improves subsequent performance. Differences across runs are also descriptive because

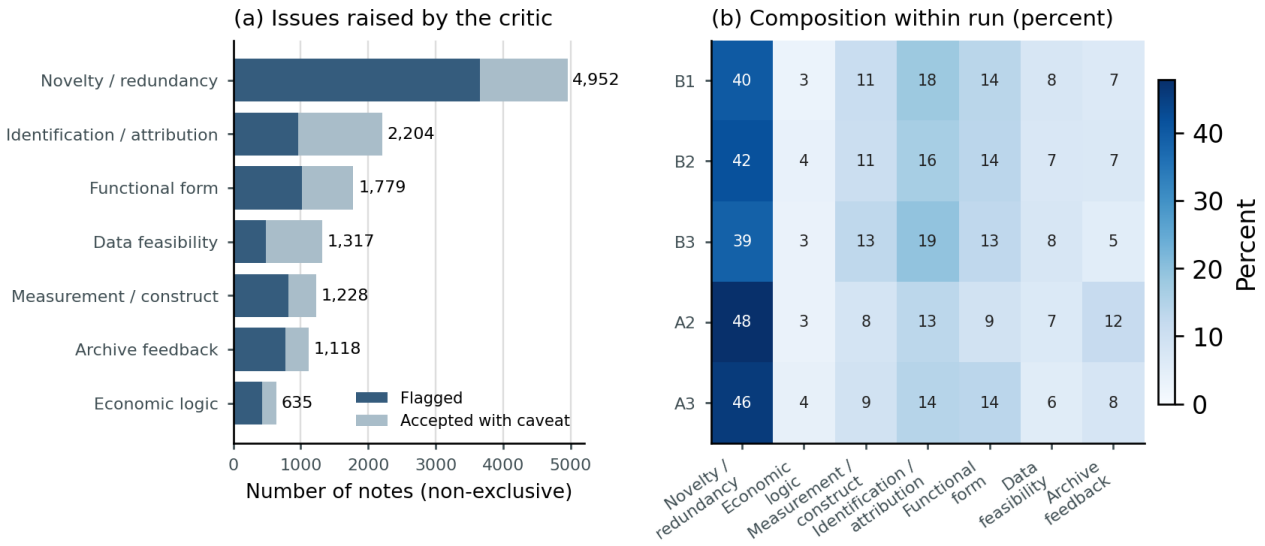


Figure 8: Objects of contemporaneous critic feedback. Panel (a) reports nonexclusive issue mentions, and Panel (b) reports the mutually exclusive dominant category among judgments containing a material objection. Coding does not use the target proposal's empirical outcome or any post-search result.

the common critic evaluates proposal streams produced by different configurations.

Figure 9 and Table 10 show how the critic operationalizes one part of this review. Among the 7,544 judgments that make an explicit novelty comparison, 39.7% classify the proposal as an archive duplicate or cosmetic variant, 14.6% as a published rediscovery, 19.5% as a limited extension, and 26.3% as a substantive new condition or combination. The critic flags 95.7% of archive duplicates and 86.6% of published rediscoveries, compared with 11.0% of limited extensions and 2.8% of substantive new combinations.

The critic's comparison set is limited to published signals invoked from its finance knowledge and proposals visible in the contemporaneous task archive; it is not an exhaustive literature review. The novelty labels therefore describe how proposals are treated within the discovery process. The conditional tests in Section 6 ask the separate question of whether a survivor retains information relative to selected predictors in a fixed external library, and likewise do not establish literature-wide novelty or validate the associated mechanism.

7.3 What the Recorded Trajectories Show—and Do Not Show

Across the cases, recorded feedback is followed by changes along three principal dimensions of a hypothesis: construct measurement, economic conditioning, and functional form. Signals 1 and 2 repair and extend a build-versus-buy construct across alternative financial-statement

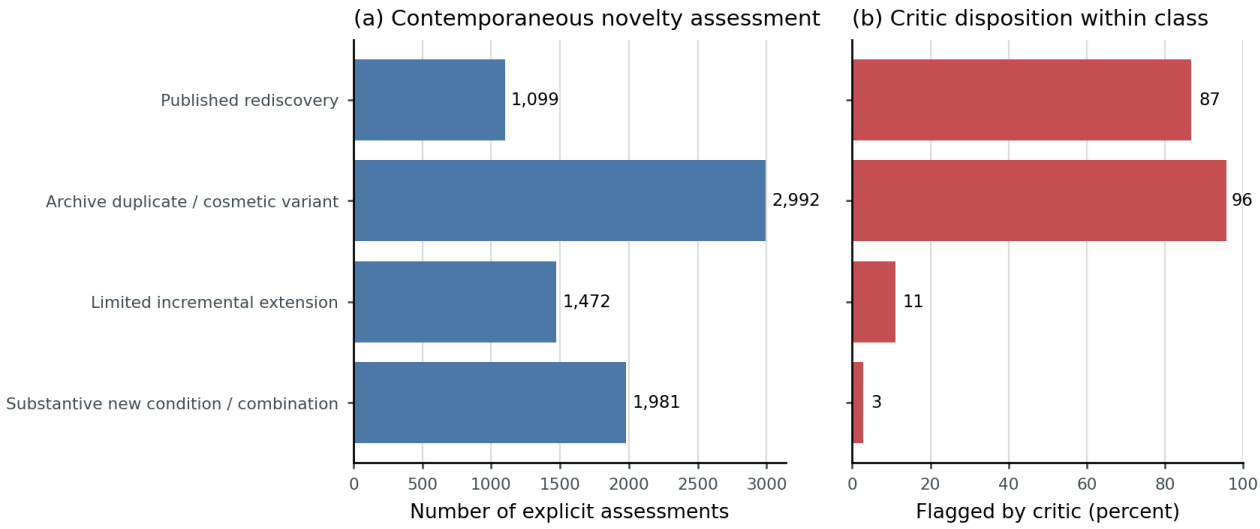


Figure 9: Contemporaneous novelty assessments in critic judgments. Panel (a) reports classifications among judgments making an explicit comparison, and Panel (b) reports the share flagged within each class. The classification and disposition come from the same judgment and therefore describe the critic's review standard rather than independently validating novelty.

representations. Signal 4 adds an operating-performance condition to an ambiguous investment pattern. Signals 6 and 10 reassign familiar inputs from compensating or stand-alone scores to binding conditions, although one follows critique and the other follows an archived empirical diagnostic. Each characterization can be linked to feedback available at the time and to a subsequent change in the executable expression.

These cases are selected for survival and legibility. They therefore do not identify how often such changes occur, their average empirical value, the causal contribution of critique, or the truth of the economic mechanisms attached to the final expressions. Nor do they establish that every component is absent from prior research. They support a narrower process claim: within this architecture, subsequent proposals sometimes respond to recorded critique and empirical feedback through systematic and auditable changes in measurement, conditioning, and functional form. The empirical pipeline determines which hypotheses survive; the trace makes the sequence of research decisions leading to those hypotheses inspectable.

8 Discussion

The evidence is best interpreted as showing what a transparent AI-assisted discovery architecture can produce and where its empirical limits become binding. The framework generates

a large population of interpretable accounting-based hypotheses, records the adaptive process that produces them, and evaluates every executable hypothesis under a common empirical protocol. The resulting funnel separates the productivity of the search from the credibility and distinctiveness of its eventual discoveries.

8.1 Search Productivity versus Validated Discovery

The central empirical pattern is the sharp gap between initial search yield and validated discovery. Each primary run evaluates 1,603–1,607 hypotheses, of which 949–1,001 clear the conventional first-pass screen in their declared direction. Within-task joint comparisons reduce this population to 73–82 horse-race survivors, and 52–68 of those also survive the run-level multiple-testing screen. Subsequent requirements for temporal stability, direction-consistent alphas under three factor models, and incremental information relative to published predictors produce substantially greater attrition. Only two signals in each primary run survive the complete protocol. High first-pass success rates therefore characterize the productivity of the search; they should not be interpreted as evidence that most generated hypotheses constitute distinct or robust return predictors.

The configuration results provide descriptive evidence about how the search evolves. First-pass performance improves after accumulated empirical feedback becomes available even in the no-critique run, suggesting that critique is not necessary for this form of within-sample adaptation. Critique is instead associated with a broader distribution of economic mechanisms and a higher yield of within-task incremental survivors relative to the number of hypotheses evaluated. Because each comparison configuration is represented by one run and the no-critique configuration also uses less model compute, these patterns do not identify separate causal effects of the archive, proposer, and critic. The three independently initialized primary runs likewise produce similar aggregate funnels and recurring accounting mechanisms, but no horse-race survivor recurs verbatim across them. Reproducibility therefore appears in broad economic content and aggregate search behavior rather than in deterministic reproduction of identical formulas.

The value of the symbolic design is process observability. Each hypothesis states an economic mechanism, an executable accounting expression, and a return direction before its own result is observed. Deterministic code then constructs and tests every validated signal, while the recorded process trail preserves unsuccessful proposals, critic notes, validation findings, re-

visions, and empirical outcomes. This structure makes it possible to examine not only which formulas predict returns, but also how the realized candidate family was constructed and how much search preceded the reported survivors. The complete-protocol survivors remain candidates for external validation rather than definitive new anomalies or causal evidence for their proposed mechanisms.

8.2 Look-Ahead Bias: Measurement Leakage versus Hypothesis Selection

The use of a pretrained LLM in a historical backtest raises a legitimate look-ahead concern. Language models can retain information encountered during pretraining, and exposure to evaluation material can distort measured performance (Carlini et al., 2021; Golchin and Surdeanu, 2024). In finance, Glasserman and Lin (2024) emphasize this concern for retrospective GPT sentiment measures: when a model reads an identified historical document, it may already possess information about the firm or the outcome that followed. More generally, credible LLM-based prediction requires careful attention to training leakage and to the statistical properties of AI-generated variables (Ludwig et al., 2026; Battaglia et al., 2024). The severity and form of this concern depend critically on where the LLM enters the empirical pipeline.

Let $\mathcal{F}_t$ denote the information available at date t. In an observation-level measurement design, an LLM might construct a historical sentiment score as

$$\widetilde{S}_{it} = M_\theta(D_{it}, i, t),$$

where D_{it} is a historical document and θ denotes parameters trained on a corpus that may contain post-t information. Even if $D_{it} \in \mathcal{F}_t$, the resulting score need not depend only on that information. A firm name, date, or event description may activate knowledge stored in θ about what subsequently occurred. Future information can therefore enter the historical regressor differently for different firm-date observations. This is an observation-level measurement-leakage channel. A model trained before the evaluation period, deidentified inputs, and prospective evaluation can reduce this channel, but it is difficult to exclude when a post-sample black-box model retrospectively scores identified documents.

Our architecture places the LLM at a different point in the empirical pipeline. The model

proposes a global symbolic expression e_j, after which fixed external code constructs

$$S_{it}^{(j)} = e_j\left(X_{it}^{\text{lag}}\right),$$

where X_{it}^{lag} denotes accounting inputs aligned under the prespecified information-timing rule. The LLM does not read firm-level observations, assign historical scores, access the raw CRSP–Compustat panel, or alter the portfolio-construction procedure. The same inspectable expression is applied to every firm and period, and accounting observations for fiscal years ending in calendar year y first become eligible in July of $y+1$. Conditional on a frozen expression, signal values are therefore determined mechanically by the accounting inputs supplied to the pipeline rather than by firm-specific knowledge contained in the model. The model can influence which global expression is tested, but it cannot use firm i's subsequent return to tailor firm i's historical signal value. This design removes the direct LLM-mediated firm-by-firm measurement channel; it does not make the selected expression prospective.

The timing rule warrants a separate qualification. A six-month reporting lag is not equivalent to using a vintage, as-originally-reported accounting database. If the Compustat extract contains subsequent restatements or database revisions, conventional accounting-data look-ahead concerns may remain even though the LLM never observes the panel. Unless the underlying data are genuinely point in time, the architecture removes observation-level *LLM measurement leakage*, conditional on the supplied accounting panel and timing conventions, rather than rendering every historical accounting input measurable with respect to $\mathcal{F}_t$.

The first remaining LLM-specific concern is pretraining-mediated hypothesis selection. Because the model's training corpus may include finance and accounting research, it could recall a published formula, a successful mechanism, or the broader historical lesson that certain classes of accounting variables predict returns. The contemporaneous critic records show that the workflow encounters and often filters proposals it classifies as familiar. Among 7,544 critic notes containing an explicit novelty comparison, 1,099 classify the proposal as a published rediscovery, and the critic flags 86.6% of those cases. These classifications describe how the critic operationalizes its novelty screen; they are not a formal test of training contamination. The critic does not conduct an exhaustive literature search, and the model may recombine remembered ideas without reproducing a published formula exactly.

The comparisons with the 209 Chen–Zimmermann predictors provide a more demanding empirical diagnostic. After the multiplicity, temporal, and factor-model gates, 103 candidates

enter the published-predictor spanning stage. Of the 101 with estimable conditional tests, 11 retain a direction-consistent conditional statistic above 1.96 and only three remain above 3.00; the absolute conditional statistic is below its univariate counterpart for every estimable candidate. For this small subset, the evidence makes simple relabeling of a nearby published predictor less plausible. It does not establish that the underlying ingredients or economic intuition were absent from the model's training corpus. Nor would low pairwise correlation alone establish novelty, because a nonlinear combination can have modest correlation with each component while remaining in their joint span. Conditional tests against nearby published predictors and standard factor models are therefore more informative than formula names or pairwise similarity alone.

The second remaining concern is adaptive search. From Generation 1 onward, later hypotheses respond to aggregate performance summaries computed from the same 1963–2024 return panel used to evaluate earlier proposals. This is deliberately an in-sample evolutionary search and falls within the data-snooping and adaptive-data-analysis problems studied by White (2000); Harvey et al. (2016); Dwork et al. (2015); Russo and Zou (2016). Recording every executable hypothesis materially improves search transparency because the realized testing family contains unsuccessful as well as successful trials, allowing multiplicity procedures to operate on roughly 1,600 recorded evaluations per primary run. Nevertheless, observing the realized denominator does not solve the entire adaptive-inference problem. Our post-search procedures condition on the realized set of executable hypotheses. They do not rerun the full proposer–critic–revision architecture under the null and therefore should not be interpreted as end-to-end selective inference for the search algorithm. Nor can they count unobserved research attempts embodied in the model's pretraining corpus.

The temporal, factor-model, and published-predictor tests should consequently be viewed as increasingly demanding robustness and spanning diagnostics, not as true out-of-sample validation. In particular, post-2010 performance is not out of sample because the agents receive feedback from the full return history and the pretrained model may have encountered research produced during that period. A true out-of-sample test would freeze the model, prompts, executable formulas, data conventions, and empirical protocol before new returns are realized and would evaluate the selected hypotheses prospectively without further revision. Look-ahead bias does not disappear. The architecture narrows and exposes its channels: it removes direct observation-by-observation LLM measurement leakage while leaving an inspectable problem of

pretraining-informed and adaptively data-informed hypothesis selection.

8.3 Generalizability and Limitations

The framework is portable to other empirical settings that share three features: a structured set of candidate primitives, a transparent evaluation pipeline, and enough data to discriminate signal from noise. Potential applications include credit risk modeling, corporate event studies, macroeconomic forecasting, and accounting measurement. In each case, the central requirement is that candidate hypotheses can be formalized in a constrained language and evaluated under a stable protocol. The present design does not accommodate signals that require unstructured text, high-frequency data, cross-firm network effects, or real-time information. Those extensions define a natural research frontier but require fundamentally different hypothesis languages and evaluation pipelines.

Several boundaries remain. The search is restricted to 66 annual Compustat variables, 32 operators, eight themes, and the prompts and model configurations used in the study; it is not an exhaustive search over accounting information. The three primary repetitions reveal stable aggregate patterns but substantial stochasticity in the exact expressions discovered. The alternative configurations are each run once and differ in model calls as well as topology, so their comparisons remain descriptive rather than causal estimates of model or critic effects. Once discovered, an executable expression and its empirical results are reproducible under the fixed pipeline, but exact reproduction of the generative search path is neither observed nor expected. Extensions to unstructured disclosures, high-frequency data, or firm networks would require new hypothesis languages and, when the LLM directly scores firm-level observations, renewed attention to the measurement-leakage channel described above.

8.4 Conclusion

We develop a human-designed, AI-executed framework for hypothesis discovery in empirical asset pricing. Across three independently initialized primary runs, the system evaluates approximately 1,600 hypotheses per run and produces similar aggregate discovery funnels despite stochastic variation in the specific expressions generated. Many hypotheses clear an initial significance screen, but the population contracts sharply when related signals compete jointly and when the survivors must pass run-level multiple-testing correction, temporal-stability requirements, three factor models, and conditional tests against 209 published predictors. Two

signals in each primary run survive the complete protocol; across all six separately evaluated runs, 11 signals do so. The central empirical lesson is therefore one of disciplined attrition: AI can make hypothesis production highly scalable, but robust and incremental accounting-based return predictability remains scarce.

The recorded discovery process shows subsequent proposals repairing accounting measures, extending mechanisms across alternative representations, adding conditions that sharpen ambiguous economic interpretations, and changing the functional role assigned to familiar variables after critique or empirical feedback. These findings do not imply a fully autonomous scientist and do not eliminate the need for prospective validation. They instead identify a productive division of labor: human researchers design the accounting measurement system and evidentiary standards, AI research agents conduct a broad and adaptive search within those boundaries, and deterministic code makes each attempt comparable and auditable. Under this architecture, AI can expand the scope of accounting hypothesis generation while making the resulting search more transparent and statistically accountable.

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Table 1: Agent Configurations

Configuration	Proposer	Critic	Interaction	Calls/gen.	Runs
Primary	`claude-opus-5` (Anthropic)	`claude-opus-5` (Anthropic)	Propose–critique–revise	2–3	3
Single-agent	`claude-opus-5` (Anthropic)	—	Propose	1	1
GPT-5.5 proposer	`gpt-5.5` (OpenAI)	`claude-opus-5` (Anthropic)	Propose–critique–revise	2–3	1
GPT-5.6-sol proposer	`gpt-5.6-sol` (OpenAI)	`claude-opus-5` (Anthropic)	Propose–critique–revise	2–3	1

Every run covers the same 28 cross-theme tasks with 10 generations per task and 6 requested proposals per generation, uses high reasoning effort for every model role, excludes financial firms and microcaps from the evaluation universe, applies the expected-sign Fama–MacBeth screen at $|t| > 1.96$, and reports the discovery-stage long–short alpha against the Fama–French five-factor model. The configurations differ only in the proposer model or in whether the critique–revision subloop is present; the single-agent configuration therefore also makes fewer model calls per generation. The primary configuration is executed in three independent runs; each comparison configuration is executed once. Each run is its own search universe and its own post-discovery testing family.

Table 2: Discovery Funnel Across Independent Runs

Run	Consid.	Reject.	Eval.	Correct	Weak	Wrong	Error	Hit (%)	Survivors
B1	1,680	73	1,607	949	615	39	4	59.1	73
B2	1,680	77	1,603	961	604	34	4	60.0	78
B3	1,680	73	1,607	1,001	568	38	0	62.3	82
A1	1,680	68	1,612	1,211	388	11	2	75.1	62
A2	1,681	70	1,612	659	913	40	0	40.9	69
A3	1,681	66	1,615	767	813	34	1	47.5	85

"Considered" counts parsed final proposals; "Reject." denotes validation rejections before empirical evaluation. Correct, weak, wrong-sign, and error outcomes partition the evaluated population. The hit rate is correct-sign first passes divided by evaluated proposals. Survivors are the final outputs of the within-run, within-pair horse races. B1–B3 are separately initialized primary runs; A1 is the no-critique comparison, and A2–A3 replace the proposer model while retaining critique. Runs are not pooled.

Table 3: Cross-Run Repeatability of the Primary Search

	Task-level rank correlation		Shared exact formulas		Shared variable sets	
Run pair	Hit rate	Survivors	Evaluated	Survivors	Evaluated	Survivors
B1–B2	0.535	0.173	3	0	128	4
B1–B3	0.502	0.392	1	0	149	5
B2–B3	0.486	-0.001	2	0	136	5

Rank correlations are Spearman's ρ across the 28 tasks, computed separately for task-level correct-sign hit rates and for task-level horse-race survivor counts. Shared exact formulas require equality after deterministic expression normalization. Shared variable sets require only that two formulas draw on the same set of accounting variables, ignoring operators and structure; this is a coarse structural proxy that sits between literal recurrence and the mechanism-level coding reported in Table 4. Distinct variable sets per run: B1: 1,012 evaluated, 72 survivor; B2: 959 evaluated, 75 survivor; B3: 1,007 evaluated, 81 survivor.

Table 4: Mechanism-Level Recurrence Across the Primary Runs

Panel A: Pair-mechanism-sign recurrence

Population	Pair-mechanisms in 3/3 runs	In exactly 2/3 runs	In 1/3 runs
Evaluated hypotheses	198	90	88
Horse-race survivors	8	38	94

Panel B: Macro-averaged pair-level Jaccard overlap

Run pair	Evaluated hypotheses	Horse-race survivors
B1–B2	0.662	0.166
B1–B3	0.697	0.205
B2–B3	0.666	0.203

A pair-mechanism is one (task, primary mechanism, predicted sign) label assigned under a codebook frozen before any post-search statistic was viewed. Coding uses the hypothesis, expression, sign, and task only. Panel A counts distinct pair-mechanism-sign labels by how many primary runs contain at least one hypothesis carrying that label. Panel B computes Jaccard overlap of the mechanism-sign sets separately within each task and then averages equally over the 28 tasks. The two independent coding passes agreed on 88.8% of records (Cohen's $\kappa = 0.887$); all disagreements were independently adjudicated. The residual Other label accounts for 1.88% of records.

Table 5: Mechanism Breadth and Concentration Across Tasks

Run	Mean count	Median count [range]	Mean HHI	Median HHI [range]
B1	7.75	8 [6, 9]	0.210	0.213 [0.136, 0.334]
B2	7.64	8 [6, 9]	0.225	0.206 [0.148, 0.324]
B3	7.57	8 [5, 9]	0.236	0.219 [0.128, 0.473]
A1	6.82	7 [4, 9]	0.334	0.298 [0.191, 0.596]
A2	7.25	7 [6, 9]	0.220	0.217 [0.150, 0.375]
A3	7.46	8 [5, 9]	0.217	0.192 [0.146, 0.372]

Statistics are computed separately for each of the 28 tasks from all evaluated hypotheses in a run. Count is the number of distinct substantive mechanism labels used in the task. HHI is $\sum_m s_m^2$, where s_m is a mechanism's share among non-Other records in that run and task; lower values denote a more even distribution across mechanisms. The residual Other label is excluded from both measures. B1–B3 are separately initialized primary runs, A1 is the no-critique comparison, and A2–A3 replace the proposer model while retaining critique.

Table 6: High-Sharpe Independent Discoveries in the Primary Runs

Run	Signal	Theme pair	Gen.	Univariate t	Conditional t	Alpha t	Sharpe
B1	Symmetric Policy And Net Issuance Composite	Accruals x Financing	9	4.82	2.39	2.65	0.65
B1	Durable Profit Per Hard Dollar Value	Valuation x Intangibles	9	6.90	2.84	5.27	0.63
B1	Burning A War Chest Raised Earlier	Accruals x Financing	4	3.07	2.38	3.37	0.62
B2	Industry Adjusted Own Peak Deferral With Burn	Accruals x Financing	9	5.35	2.35	4.23	0.69
B2	Pre Investment Cash Plus Composition Rank	Profitability x Intangibles	9	3.61	3.82	4.15	0.59
B2	Total Funding Requirement Burden	Investment x Distress	3	2.97	2.81	3.37	0.55
B3	Normalized firm level yield with wide mix	Valuation x Financing	7	6.67	6.97	4.45	0.58
B3	Distributions exceeding internal cash generation before maturities	Financing x Distress	9	3.45	2.86	3.61	0.56
B3	Liquid Assets Plus Normalized Earnings Yield Composite	Profitability x Valuation	9	5.29	3.62	5.94	0.55

For each primary run, the table reports the three final horse-race survivors with the largest sign-oriented annualized Sharpe ratios among survivors whose Fama–French five-factor alpha has the predicted sign. Univariate t is the discovery-stage Fama–MacBeth statistic; conditional t is the statistic from the final within-pair stepwise regression. All statistics are sign-oriented using the direction declared before evaluation. Appendix Table B.6 lists the expressions. This ranking is descriptive and does not replace search-universe inference.

Table 7: Cumulative Post-Search Attrition Within the Horse-Race Set

Run	Horse	BY	Bayes	Double boot.	Subsamples	Non-decay	Alpha 3/3	CZ
B1	73	59	59	59	59	38	19	2
B2	78	52	52	52	52	37	18	2
B3	82	68	68	68	68	45	23	2
A1	62	54	54	54	54	31	21	4
A2	69	45	45	45	45	27	11	1
A3	85	47	47	47	47	29	11	0
Total	449	325	325	325	325	207	103	11

Counts are cumulative intersections within each run's final within-pair horse-race set. BY is the run-level false-discovery-rate screen; Bayes requires posterior null probability below 5% under the moderate prior; double bootstrap requires the run-specific 5% hurdle; subsamples requires significance in at least four of seven windows; non-decay excludes signals classified as decaying; Alpha 3/3 requires direction-consistent significance under FF5, FF6, and the q-factor model; CZ requires a direction-consistent conditional statistic above 1.96 against the selected Chen–Zimmermann controls. The six runs remain separate testing families; the total row is descriptive and does not pool their multiple-testing calculations. WLS is a reported diagnostic, not a gate.

Table 8: Signals Surviving the Complete Post-Search Protocol

Run	Signal	Theme pair	Gen.	FMB t	Horse t	Robust	Min. $\lvert t^{\alpha}\rvert$	CZ t	CZ mo.	VW t
B1	purchased share of growth spending	Profitability x Investment	8	5.12	3.63	6/7	3.68	2.47	27	3.34
B1	knowledge capital built versus bought	Profitability x Investment	9	4.30	2.60	5/7	4.35	2.40	24	3.34
B2	Predictable Accruals Sustained Profit	Accruals x Quality	7	4.02	3.45	7/7	2.25	1.98	603	1.27
B2	Funded Research Tilt With Profit Delivery	Investment x Intangibles	7	5.15	4.33	7/7	3.92	3.23	24	3.94
B3	Working Capital Bloat Vs Peers	Accruals x Quality	2	5.47	2.67	7/7	3.74	2.18	737	2.99
B3	Conservatism Adjusted Profit Lean Claims	Profitability x Accruals	3	4.71	2.57	7/7	3.22	2.93	587	2.98
A1	Sustained Coverage With Capped Cushion	Profitability x Distress	9	3.77	3.93	6/7	2.96	2.89	737	1.69
A1	Broad Universe Profitable Value Model	Profitability x Valuation	9	5.88	5.45	6/7	4.16	2.87	636	3.94
A1	Negative Working Capital Profit Sort	Quality x Distress	1	3.12	2.20	5/7	2.66	2.47	28	0.43
A1	Trough Coverage Clean Accrual Rollover	Quality x Distress	5	5.51	3.62	7/7	2.97	3.17	437	2.46
A2	Gross Margin Inventory Efficiency	Profitability x Investment	1	3.97	2.97	6/7	2.67	3.38	737	0.89

These 11 signals pass every cumulative gate in Table 7. Robust is the number of significant windows out of seven. Min. $\lvert t^{\alpha}\rvert$ is the smallest absolute alpha statistic across FF5, FF6, and the q-factor model, all of which have the declared direction. CZ t is the conditional Fama–MacBeth statistic after controlling for the selected published predictors, and CZ mo. is the number of return-valid monthly cross sections available for that conditional test. VW t is the market-equity-weighted Fama–MacBeth statistic, a sensitivity diagnostic rather than a survival gate; seven of the eleven signals exceed 1.96 in the declared direction. Appendix Table B.9 reports the executable expressions. Signal numbers used in Section 7 follow this row order.

Table 9: Taxonomy of Contemporaneous Critic Objections

Dominant object	Total	Share (%)	B1	B2	B3	A2	A3
Novelty / redundancy	3,473	42.9	647	679	627	779	741
Identification / attribution	1,321	16.3	290	268	313	218	232
Functional form	1,017	12.6	221	220	208	147	221
Measurement / construct	844	10.4	178	175	209	136	146
Archive feedback	619	7.6	105	111	84	190	129
Data feasibility	563	6.9	123	116	126	108	90
Economic logic	264	3.3	46	57	49	49	63

The table classifies the dominant object of criticism in contemporaneous critic notes that contain a material objection. Categories are mutually exclusive. "Archive feedback" identifies failure to use a diagnostic already visible within the same task; simple formula similarity is coded as novelty/redundancy. No evaluation outcome for the criticized proposal, survivor status, or post-search result enters the coding corpus.

Table 10: Critic Assessments of Proposal Novelty

Assessment	Count	Share (%)	Flagged (%)
Published rediscovery	1,099	14.6	86.6
Archive duplicate / cosmetic variant	2,992	39.7	95.7
Limited incremental extension	1,472	19.5	11.0
Substantive new condition / combination	1,981	26.3	2.8
Novelty not discussed	854	10.2	73.5

Shares for the first four rows use only notes that make an explicit novelty comparison; the residual row reports its share of all critic notes. "Flagged" is the critic's contemporaneous disposition, not an empirical outcome. The comparisons are limited to published signals invoked by the critic and the visible within-task archive; they are not exhaustive literature classifications.

Appendix A Implementation and Measurement Details

This appendix documents the implementation conventions required to reconstruct the discovery runs and post-search analysis. Throughout, a *proposal* is a structured model output before deterministic validation; a *hypothesis* is the mechanism–expression–direction triplet $h_j = (m_j, e_j, d_j)$; an *expression* or *formula* is the executable component e_j; and a *signal* is the resulting firm-period quantity $S_{it}^{(j)}$. We use *survivor* only with the relevant stage of the empirical funnel, culminating in the *complete-protocol survivors* defined in Section 6.

A.1 Hypothesis Language, Task Contract, and Validation

The discovery language contains 66 annual accounting variables: 33 balance-sheet items, 16 income-statement items, and 17 cash-flow or financing items. Each entry in the frozen catalog records its Compustat mnemonic, statement category, economic description, theme memberships, whether it is shared across theme catalogs, common scaling variables, and any variable-level directional annotation supplied to the agents. The language also contains 32 operators in five classes: scaling and change, time-series state, composition, mathematical transformation, and cross-sectional or industry-relative operations. Table B.1 reports their executable signatures and economic roles; the frozen operator catalog records the exact parameter domains.

Every pair-level job begins from a serialized task contract. The contract specifies the two themes, their economic objective, conceptual examples of possible cross-theme mechanisms, theme-specific variable resources, admissible operators, the empirical evaluation protocol, and duplicate rules. Table B.2 reports the objective for each of the 28 unordered theme pairs. The examples are conceptual anchors, not executable templates or formulas identified as empirically successful.

A structured proposal contains six required fields: a unique name, an executable expression, a stated economic mechanism, a declared return direction, construction-type metadata, and a concise rationale. The construction-type field is labeled `interaction_type` in the output schema presented to the agents; it covers additive, multiplicative, conditional, and weakest-link constructions and does not imply an econometric interaction.

Before empirical evaluation, deterministic validation checks the required fields, declared direction, construction type, balanced syntax, operator membership and arity, parameter domains, variable membership, and duplicate rules. It also requires an expression to contain at

least one variable assigned exclusively to each theme in its task. A variable shared across the two theme catalogs, including a generic denominator, cannot satisfy both requirements. This is a syntactic rule based on the prespecified catalogs, not a claim that an accounting variable has only one economic interpretation. Duplicate checks operate both within the current batch and against exact or normalized expressions in the prior task archive. The arity check is repeated at evaluation time. A proposal that fails validation remains in the recorded process trail but does not become an empirically evaluated hypothesis.

A.2 Generation-Loop Realization and Model Context

Each complete search run is resolved into 28 operationally isolated pair-level jobs. A pair-level job has its own task contract, proposer state, evaluated archive, validation history, budget, and record set. No archive or conversational state is shared across theme pairs or runs. The resolved configuration and serialized task contract are written before the first model call.

In the proposer–critic topology, one pair-specific proposer session persists across generations, whereas a fresh critic session is created for each generation. The proposer first returns a batch of structured proposals. The critic observes that batch, the task contract, and the archive of earlier evaluations, but not the current batch's empirical outcomes. It returns a verdict, score, and rationale for each proposal. If at least one proposal is flagged, the same proposer session receives the critic feedback and revises the flagged subset. A pair-generation therefore uses two model calls when no revision occurs and three when revision occurs. The no-critique configuration uses one proposer call per pair-generation.

Generation 0 begins with the task objective and economic story, theme resources, conceptual anchors, operator and parameter rules, definitions of the empirical summaries, the required output schema, and the generation budget. Beginning with Generation 1—the second generation—the proposer also receives the complete evaluated archive for its pair-level task. Each archive entry reports the expression, declared direction, Fama–MacBeth statistic, long–short alpha and alpha statistic, Sharpe ratio, coverage, and any execution error. Recent critic, parser, and validation feedback is also supplied. A revision call receives the current batch, the proposal-specific critic objections, and the same expression and output constraints. Neither the proposer nor the critic receives firm-level CRSP or Compustat observations or firm-level future returns.

The primary runs B1–B3 use `claude-opus-5` as both proposer and critic. A1 retains

`claude-opus-5` as proposer but removes critique and revision. A2 and A3 use `gpt-5.5` and `gpt-5.6-sol`, respectively, as proposers while retaining `claude-opus-5` as critic. All roles use the high-reasoning setting available from their respective providers. These provider-specific settings do not define a common compute unit. The design uses neither fine-tuning nor retrieval augmentation.

Calls are issued through each provider's standard message interface. Every role in every run uses the high reasoning setting, a sampling temperature of 1.0, no nucleus-sampling restriction, and a limit of 16,000 tokens on generated output. Calls to the Anthropic interface request adaptive rather than budgeted deliberation, so no explicit deliberation-token budget applies. No provider-side random seed is set and none is available through either interface, so run-to-run variation in model output is not controllable by the design. The output limit does not bind: the longest of the 3,919 recorded responses contains 14,408 tokens. The six runs were executed between 3 and 6 August 2026, and the data-construction, evaluation, and model-interface components were identical across all of them.

Conversational continuity is maintained through provider-native session state. We retain prompts, visible model responses, critic judgments, and any summary or deliberative text explicitly returned through the provider interface. These are observable process artifacts; they are not treated as an exhaustive or necessarily faithful representation of latent cognition. The process analysis in Section 7 relies on observable proposals, rationales, critic judgments, revisions, executable expressions, and empirical outcomes.

Accepted proposals and revised versions of flagged proposals form the final parsed batch submitted to deterministic validation. Each such record is a *considered proposal*. A flagged original that is superseded by a revision remains recoverable from the raw proposer response, critic record, and rendered revision prompt, but is not counted as an additional considered proposal or empirical test. An unparseable record never enters the considered population. A considered proposal that passes deterministic validation is a *validated hypothesis*; every validated hypothesis submitted to the grader is an *empirically evaluated hypothesis*, including one that subsequently ends in an execution error.

Each pair-level job completes the fixed schedule of 10 generations and requests six proposals per generation. A complete run therefore contains 60 requested slots per task, 280 pair-generations, and 1,680 requested proposal slots. A model response can return more or fewer parseable final records than requested. A3 returns one additional parsed record. A2

returns two, one of which reproduces another record exactly and is removed by expression-level deduplication. Both configurations therefore reach 1,681 considered proposals rather than 1,680.

A.3 Accounting Data and Annual Signal Construction

The source histories begin in 1960, providing observations needed for lagged and rolling constructions before the return-prediction sample begins in July 1963. The CRSP panel contains ordinary common shares with share codes 10 and 11 listed on NYSE, AMEX, or NASDAQ. From these records we construct monthly returns, market equity, NYSE membership, and the security panel. Market equity is the absolute value of price multiplied by shares outstanding. Available delisting returns are compounded with ordinary monthly returns. When a delisting return is missing and the delisting code lies between 200 and 399, -30% is substituted before compounding; a missing delisting return outside that code range receives no substitute. Within a delisting month a missing ordinary return is treated as zero, so the month's return is the delisting return alone. Delisting returns adjust the monthly observation a security already has in its delisting month and are never appended as an additional observation, so a security with no monthly record in that month contributes no delisting return. This is more conservative than implementations that insert the delisting month.

The Compustat panel uses annual industrial-format, standardized, domestic, consolidated records. Compustat observations are linked to CRSP through CCM link types LC and LU and primary-link designations P and C, with link dates enforced. When several records map to the same security and fiscal year, the record with the latest fiscal period-end date is retained. An observation for a fiscal year ending in calendar year y receives formation year $y + 1$.

Each validated expression is evaluated on the firm-year panel by recursive descent over its syntax tree. Annual lags are gap-aware: if the requested calendar year is absent for a security, the lag is missing rather than taken from the nearest available observation. This convention propagates to growth, changes, moving windows, trends, volatility, acceleration, cumulative flows, and within-firm ranks. Missing accounting values are altered only when the expression explicitly invokes an admissible operator such as `COALESCE`. After the complete expression has been evaluated, its nonmissing values are winsorized within formation-year cross sections at the 1st and 99th percentiles.

Industry-relative operators are evaluated within groups defined by two-digit SIC code and

by the fiscal year underlying each formation year. The classification is the Compustat historical SIC code for that fiscal year and, where that field is missing, the CRSP classification recorded in June of the formation year. The historical Compustat code is missing for 46.9% of annual records and for every record before 1980, so the earliest part of the sample is classified entirely on the June convention and the industry membership date is not uniform across the sample.

`IND_ADJ` subtracts the contemporaneous industry median. `IND_ZSCORE` standardizes by the group mean and sample standard deviation, both computed on unwinsorized values because winsorization applies to the completed expression. `IND_RANK` assigns within-group percentile ranks, with ties receiving their average rank. No minimum group size is imposed. A group with a single member or with no dispersion returns a missing value from `IND_ZSCORE` rather than falling back to the pooled cross section, and a security with no industry assignment is missing for all three operators. Such securities are not removed from the monthly panel; they receive the cross-sectional median rank under the filling rule described below.

Annual values are mapped to monthly CRSP observations using the July–June convention. An accounting observation for a fiscal year ending in calendar year y first enters the return-prediction panel in July of $y + 1$ and remains active through June of $y + 2$. July through December use the current calendar formation year, whereas January through June use the preceding formation year.

This convention imposes a minimum six-month lag between fiscal year-end and first portfolio formation. It does not establish that the Compustat extract is a vintage, as-originally-reported database. If the extract incorporates subsequent restatements or database revisions, conventional accounting-data look-ahead concerns can remain even though the LLM never reads the firm-level panel.

A.4 Monthly Sample and Deterministic Evaluation

Evaluator inputs are normalized to calendar month-end and reindexed to a contiguous return sample from July 1963 through December 2024. Before monthly missing-value filling or investable-universe exclusions, raw coverage is computed on the aligned security-month signal as the share of nonmissing values in the complete month-by-security rectangle of the evaluation sample, that is, 738 months and 26,264 securities, or 19,382,832 security-months. The denominator spans the entire sample period and the entire cross section rather than the interval over which a given security or signal is active, so no expression can approach full coverage under it:

across the 9,656 evaluated hypotheses, realized coverage has a median of 0.111 and a maximum of 0.149. An expression with less than 1% nonmissing coverage is rejected, which 11 hypotheses across the six runs fail. This threshold is a mechanical executability screen, not evidence of adequate coverage in the final investable sample or in later joint regressions.

Within an active signal month, nonmissing values are converted to percentile ranks centered at zero, with support approximately $[-0.5, 0.5]$. A security with observed market equity but a missing signal receives the cross-sectional median rank of zero. Entire months before the first valid signal observation or after the last valid observation remain missing, so the filling rule does not extend the signal's active date range.

Financial-firm and microcap filters use formation-month information. Financial firms are identified using CRSP SIC codes 6000–6999. The microcap boundary is the 10th percentile of NYSE market equity in the formation month. NYSE, AMEX, and NASDAQ securities above this boundary remain eligible, whereas NYSE securities alone determine portfolio breakpoints. Ineligible observations are removed from the characteristic and formation-weight panels. The one-month characteristic lag relates month-t eligibility and signal information to month-$t+1$ returns.

The portfolio evaluator uses NYSE signal-quintile breakpoints, value weights based on formation-month market equity, and a primitive Q5-minus-Q1 return. It reports the annualized long–short alpha, the corresponding alpha statistic, and the annualized Sharpe ratio. The discovery-stage alpha uses the Fama–French five-factor model. Unless explicitly labeled raw, portfolio statistics reported relative to hypothesis j are multiplied by its declared direction d_j, so a positive value denotes agreement with the prediction.

The cross-sectional evaluator estimates the monthly regression

$$R_{i,t+1} = a_{t+1} + b_{t+1}^{(j)} z_{i,t}^{(j)} + \varepsilon_{i,t+1}^{(j)},$$

where $z_{i,t}^{(j)}$ is the centered rank of signal j in formation month t. It reports the time-series mean of the monthly slopes and its Newey–West statistic using six lags. The archived empirical record contains the Fama–MacBeth statistic, long–short alpha and alpha statistic, Sharpe ratio, coverage, and any execution error.

An empirically evaluated hypothesis is a first-pass success when its statistic is finite and

$$d_j t_{\mathrm{FMB},j} > 1.96.$$

The first-pass success rate divides the number of such hypotheses by all hypotheses submitted to empirical evaluation in the relevant run, generation, or task, including submissions ending in execution errors. This rule is a discovery-stage screen rather than multiplicity-adjusted inference.

A.5 Parallel Execution, State, and Provenance

Pair-level jobs are executed independently and may run concurrently, so model calls from different tasks overlap. Empirical evaluation is deterministic and invariant to execution order: results do not depend on how many jobs run in parallel, and within each generation proposals are returned in proposal-input order.

Each pair-level job has a self-contained archive, and a run-level index records the 28 pair-level archives constituting a complete run. The retained artifacts include the resolved configuration, task contract, prompts, visible model responses, provider-returned summaries where available, final parsed and superseded proposals, critic judgments, validation findings, empirical evaluations, cumulative archive, diagnostics, and status records. The archive is checkpointed after every generation.

Post-search stages use an immutable hypothesis-record identifier constructed from run and task provenance together with the proposal name and normalized expression. Human-readable names are never sufficient join keys. This identity is verified against the discovery archive and horse-race record before a signal enters any post-search stage. The identifier links each reported statistic to the originating proposal, model-call record, validation decision, discovery-stage evaluation, within-task selection record, and post-search history.

A.6 Post-Search Selection, Inference, and Robustness

Post-search procedures are indexed to distinct empirical populations. All empirically evaluated hypotheses with usable full-sample Fama–MacBeth statistics in a run form that run's multiplicity family. The signed double bootstrap uses the corresponding executable hypotheses for which the monthly slope series can be reconstructed. Correct-sign first-pass hypotheses compete within their assigned theme-pair task, and the resulting horse-race survivors enter the cumulative temporal, factor-model, and published-predictor tests. Each run remains a separate testing family; no adjustment, bootstrap calibration, or selection threshold is estimated from a pooled cross-run population.

Let t_j denote hypothesis j's full-sample Fama–MacBeth statistic and $d_j \in \{-1, 1\}$ its declared direction. Define the sign-oriented statistic as

$$\widetilde{t}_j = d_j t_j,$$

so that a positive value supports the prediction.

Run-level multiplicity procedures. Let M_r denote the number of usable full-sample statistics in run r. The counts are 1,603, 1,599, 1,607, 1,610, 1,612, and 1,614 for B1, B2, B3, A1, A2, and A3, respectively. Thus, 9,645 of the 9,656 hypotheses submitted to empirical evaluation yield usable statistics for multiplicity analysis.

Let p_j denote the raw reference p-value supplied to the multiplicity procedures,

$$p_j = 2\left(1 - \Phi(|t_{\mathrm{FMB},j}|)\right),$$

where Φ is the standard normal distribution function. BY, Bonferroni, and Holm all use this same two-sided convention. With ordered values $p_{(1)} \le \cdots \le p_{(M_r)}$ and

$$c_{M_r} = \sum_{\ell=1}^{M_r} \frac{1}{\ell},$$

the Benjamini–Yekutieli (BY) procedure identifies the largest k satisfying

$$p_{(k)} \le \frac{k}{M_r} \frac{q}{c_{M_r}}, \qquad q = 0.05,$$

and rejects the corresponding k null hypotheses. Bonferroni and Holm family-wise-error adjustments are computed from the same run-level family and are sensitivity diagnostics, not gates in the complete-protocol classification. Appendix Table B.8 reports the run-level thresholds and rejection counts for all three procedures.

Because p_j depends on $t_{\mathrm{FMB},j}$ only through its magnitude, none of the three procedures conditions on the declared direction, and between 6 and 17 of each run's BY rejections carry a statistic significant opposite to the declared direction. The declared-direction requirement is imposed separately, at horse-race selection, so every record reaching the complete-protocol classification has already satisfied it.

Minimum-Bayes-factor screen. Following Harvey (2017), we convert the magnitude of the Fama–MacBeth statistic to the minimum Bayes factor

$$\mathrm{MBF}_j = \exp\left(-\frac{t_j^2}{2}\right).$$

The moderate prior assigns probability $\pi_1 = 0.20$ to a nonzero effect and $1 - \pi_1 = 0.80$ to the null. The resulting posterior-null bound is

$$\Pr(H_{0j} \mid t_j) = \frac{\mathrm{MBF}_j(1-\pi_1)/\pi_1}{1 + \mathrm{MBF}_j(1-\pi_1)/\pi_1} = \frac{4\,\mathrm{MBF}_j}{1 + 4\,\mathrm{MBF}_j}.$$

A hypothesis passes when this quantity is below 0.05, which is equivalent to $|t_j| > 2.943$. Because the minimum Bayes factor is optimized over alternatives rather than derived from a fully specified effect-size distribution, this is a minimum-Bayes-factor posterior bound, not a posterior under an explicitly parameterized slab prior. The declared-direction condition is inherited from horse-race selection.

Signed double bootstrap. The signed double bootstrap is calibrated separately for each run using its reconstructed monthly Fama–MacBeth slope matrix (Harvey and Liu, 2020). Columns corresponding to declared-negative hypotheses are multiplied by -1 before calibration, so positive statistics consistently support the declared direction. The calibration assumes a 10% non-null share, uses 100 ranking bootstraps and 200 evaluation bootstraps within each ranking draw, and recomputes Newey–West statistics with six lags. It evaluates candidate hurdles on the grid

$$1.0, 1.1, \ldots, 5.0$$

and selects, separately for each run, the smallest hurdle whose estimated false-discovery rate does not exceed 5%. The estimated false-discovery rate ranges from 5.46% to 5.99% at a hurdle of 2.4 and from 4.27% to 4.70% at 2.5. Consequently, all six run-specific cutoffs equal 2.50 after discretization. A horse-race survivor clears this gate when $|t_{\mathrm{FMB},j}| \geq 2.50$. Horse-race selection has already imposed the declared direction and required the horse-race and full-sample statistics to agree in sign, so on the population to which this gate applies the magnitude condition and the corresponding signed condition coincide.

Both resampling stages draw calendar months with replacement; hypotheses are never resampled, so contemporaneous dependence across the search population is preserved. Within a

ranking draw, columns designated null are recentered to zero and columns designated non-null are recentered to their means in that draw. Missing monthly slopes remain missing and are skipped by the Newey–West estimator. The calibration is deterministic, using the integer seeds 0 through 99 for the ranking draws and generating each draw's evaluation resamples from the same stream. Because months are resampled before the Newey–West statistic is formed, its six lags relate adjacent rows of the resampled matrix rather than adjacent calendar months.

Within-task horse races. A hypothesis enters the correct-sign first pass when $\widetilde{t}_j > 1.96$. Within each run and theme-pair task, the procedure first computes average monthly cross-sectional Spearman correlations among the resulting signals. If two signals have an absolute average correlation above 0.95, it retains the one with the larger sign-oriented univariate statistic. The remaining signals enter a joint monthly Fama–MacBeth regression. Backward elimination removes the weakest signal and re-estimates the regression until every retained signal has an estimable conditional statistic above 1.96 in its declared direction. The union of the 28 pair-specific terminal sets forms the run-level horse-race population. This stage identifies incremental conditional association relative to other hypotheses generated for the same task; it does not establish statistical independence, multiplicity-adjusted significance, or novelty relative to the broader literature.

Temporal robustness and decay. The temporal suite uses seven fixed windows: July 1963–December 2024; July 1963–December 1990; January 1991–December 2024; July 1963–June 2000; July 2000–December 2024; January 2011–December 2024; and all sample months outside NBER recessions. These are common calendar windows, not signal-specific pre- and post-publication samples. Within each window, the pipeline re-estimates the univariate Fama–MacBeth regression and its Newey–West(6) statistic. A hypothesis clears the window requirement when $d_j t_{\mathrm{FMB},j} > 1.96$ in at least four of the seven windows.

The decay analysis uses 60-month rolling Fama–MacBeth regressions advancing in 12-month increments. This yields 57 windows, the first covering July 1963 through June 1968 and the last July 2019 through June 2024; each window is indexed by its center month, and the final six months of the sample fall in no window. The analysis regresses the absolute rolling statistic on an ordered time index, so it describes the evolution of estimated strength rather than of signed performance, and a hypothesis whose statistic changes sign within the sample is assessed on magnitude alone.

Consecutive windows share 48 of their 60 months, so the trend statistic uses Newey–West standard errors with four lags, the number of subsequent windows that overlap any given window. A hypothesis is classified as declining when the trend statistic is below -1.96, strengthening when it is above 1.96, and stable otherwise. Windows that are not estimable are dropped while the time index retains its original spacing, and a classification requires at least 15 estimable windows. All 325 records entering this stage receive a classification, so the insufficient-data rule does not affect the reported funnel.

The non-recession window excludes the nine NBER peak-to-trough episodes between April 1960 and the February–April 2020 contraction, held as fixed month pairs rather than drawn from a dated vintage of the published series.

Because the seven windows overlap and the agents observe feedback from the full historical panel, neither the four-of-seven rule nor the rolling classification is a prospective out-of-sample test. Both are historical robustness diagnostics.

Factor-model tests and weighted-regression diagnostic. For each temporally admissible horse-race survivor, the portfolio evaluator estimates the sign-oriented value-weighted Q5-minus-Q1 alpha under the Fama–French five-factor model, the six-factor model that augments it with momentum, and the Hou–Xue–Zhang q-factor model (Fama and French, 2015; Hou et al., 2015). A hypothesis clears the factor gate only when $d_j t_{\alpha,j} > 1.96$ under all three models. A non-estimable required alpha is treated as a failure.

As a separate sensitivity diagnostic, the monthly cross-sectional regression is re-estimated by weighted least squares using formation-month market equity as the observation weight. The time-series mean of the monthly slopes and its Newey–West(6) statistic form the market-equity-weighted Fama–MacBeth estimate. This diagnostic is reported alongside the complete-protocol results but is not a survival gate.

All three alpha statistics use Newey–West standard errors with six lags and Bartlett weights and no small-sample degrees-of-freedom correction, matching the convention used for the Fama–MacBeth statistics. Autocovariances are formed by calendar position, so months dropped for missing data do not become adjacent.

Conditional tests against published predictors. The final stage compares each candidate with the 209 predictors in the Chen–Zimmermann library (Chen and Zimmermann, 2022). Published predictors are ordered by the absolute value of their average monthly cross-sectional

Spearman correlation with the candidate. Beginning with the closest predictor, the algorithm greedily admits controls until ten have been selected or no further admissible control remains. A tentative control set must preserve characteristic-complete coverage, nonzero regressor variance, full column rank, acceptable numerical conditioning, and at least 24 characteristic-valid monthly cross sections.

After the controls are fixed, the pipeline merges the candidate, controls, and next-month returns and retains the return-complete common sample. It estimates monthly cross-sectional regressions of returns on the candidate and selected controls and computes the candidate's conditional Fama–MacBeth statistic. The univariate statistic reported alongside it is the candidate's own full-sample estimate rather than a re-estimate restricted to the conditional common sample, so where the two samples differ the movement from the univariate to the conditional statistic reflects both the added controls and the shorter sample. The candidate passes only if its conditional statistic exceeds 1.96 in the declared direction and at least 24 return-valid monthly regressions are available. A non-estimable regression fails the protocol. Two records have 24 characteristic-valid months during control selection but only 23 return-complete months once returns enter, and therefore fail this requirement.

Numerical admissibility is assessed on the correlation-scaled cross-product matrix of each monthly design, so the criterion does not depend on the units of the regressors: a design is rejected when it fails a singular-value rank test or when its scaled condition number exceeds 10^{12}. Each monthly cross section must contain at least two more jointly observed securities than regressors, which is 13 securities when the full complement of ten controls is retained. The ordering correlations are computed on every twelfth month; a published predictor is compared only when it shares at least six sampled months and 50 securities with the candidate, and a sampled month contributes only when at least 30 securities are jointly observed. Ranks are taken within each variable's own observed values before the pairwise intersection is formed, so the ordering statistic is a correlation of marginal ranks rather than a rank correlation recomputed on the common sample. When a control fails an admissibility condition the algorithm records the failure and continues down the ordering rather than stopping, so an inadmissible predictor does not truncate the control set. Predictors with exactly equal absolute correlations are not ordered by any further rule. The conditional statistic uses Newey–West standard errors with six lags.

Complete-protocol classification. Following horse-race selection, a record must clear seven cumulative gates: (1) the run-level BY decision; (2) the posterior-null-probability criterion; (3) the run-specific signed-double-bootstrap hurdle; (4) declared-direction significance in at least four of seven windows; (5) classification as stable or strengthening rather than declining; (6) declared-direction alpha significance under all three factor models; and (7) the conditional test against selected Chen–Zimmermann predictors. Failure at one gate cannot be offset by strength at another. A non-estimable required factor-model or published-predictor regression is a failure. Market-equity-weighted Fama–MacBeth estimates remain diagnostics rather than gates.

The run-family decisions in the first three procedures are computed from their full applicable populations and then mapped to the 449 run-specific horse-race records; they are not estimated only from the horse-race winners. The cumulative funnel is

$$449 \longrightarrow 325 \longrightarrow 207 \longrightarrow 103 \longrightarrow 11,$$

where the successive reported intersections correspond to BY, non-decay, all three factor alphas, and the published-predictor conditional test. Conditional on the 325 horse-race records that clear BY, the posterior-null, signed-double-bootstrap, and four-of-seven requirements are nonbinding, although they evaluate distinct threats.

These procedures condition on the realized family of executable hypotheses. They do not rerun the proposer–critic–revision architecture under the null and therefore do not constitute end-to-end selective inference for the adaptive search algorithm.

A.7 Discovery-Result Accounting

Table 2 is constructed from the frozen run artifacts. A requested slot is one position in the six-proposal batch requested for a pair-generation. A considered proposal is a parsed final proposal after any critique and revision. A validated hypothesis passes the deterministic checks, and an empirically evaluated hypothesis is submitted to the grader. An evaluated hypothesis can end in an execution error and therefore need not produce a usable statistic.

Considered-proposal counts need not equal the 1,680 requested slots because a structured response can return more or fewer parseable final records. B1–B3 and A1 each contain 1,680 considered proposals, whereas A2 and A3 each contain 1,681. Across the six runs, 10,082 considered proposals produce 9,656 empirical evaluations, including 11 execution errors; 9,645 yield

usable full-sample statistics. The evaluated/usable counts are 1,607/1,603 for B1, 1,603/1,599 for B2, 1,607/1,607 for B3, 1,612/1,610 for A1, 1,612/1,612 for A2, and 1,615/1,614 for A3. A2 records 70 validation rejections against 1,681 considered proposals and 1,612 evaluated hypotheses. In A2, one pair-generation returned two identical final records: expression-level deduplication removes the duplicate from the considered population, while the validation trail records it as a rejection. This is the only such record across the six runs.

The empirically evaluated population is partitioned exhaustively into correct-sign first passes, weak results, wrong-sign significant results, and execution errors. Rejected records remain in the validation trail but do not enter this denominator. Failure to use information assigned exclusively to both themes is the dominant validation finding, accounting for 64–77 rejections per run. The remaining findings involve invalid expressions or metadata and a small number of duplicates. Generation-level first-pass success rates in Figure 2 divide correct-sign first passes by all hypotheses submitted to empirical evaluation in the relevant generation, including execution errors.

Table B.3 reports the primary-run task cells underlying Figure 3. The three first-pass-rate columns keep B1–B3 separate; the median and brackets summarize their center and full range without pooling proposals. Survivor counts are taken from the terminal set of each task-specific horse race. The three primary runs are separately initialized search realizations, not statistically independent samples, because they use the same historical panel.

The repeatability calculations in Table 3 compare the 28 task-level first-pass rates using Spearman rank correlation. Exact-expression overlap requires equality after lowercasing, whitespace normalization, and deterministic normalization of parentheses and commas. It does not merge algebraically equivalent expressions or economically related hypotheses. Variable-set overlap ignores operators and expression structure and therefore provides a deliberately coarser comparison.

Table B.7 reports model-call quantities recorded in the run archives. Every run contains 280 initial proposer calls, one per pair-generation. Proposer–critic runs add 280 critic calls and one revision call whenever at least one proposal is flagged. B1, B2, A2, and A3 therefore contain 840 total calls. B3 contains 839 because one pair-generation requires no revision, and A1 contains 280 proposer calls. Across B1–B3, revision occurs in 839 of 840 pair-generations. Flag shares use the 1,680 requested slots as their denominator. Provider-reported output-token totals describe realized process scale rather than a common compute unit because tokenization

and cache accounting differ across providers.

A.8 Outcome-Blind Mechanism-Coding Audit

Mechanism coding is outcome-blind but not expression-blind. A deterministic export reads all 9,656 evaluated hypotheses across the six runs and writes only the run and task provenance, generation, stored proposal name, expression, declared direction, and stated mechanism. A record identifier is assigned to the frozen export itself, in export order, and serves to hold the two coding passes and the adjudication pass on the same record. We verify, before and after the export is written, that no empirical statistic, coverage measure, execution status, critic disposition, survivor status, provider-returned rationale, or post-search result is included.

We construct one codebook for each of the 28 tasks from two sources: the task's prewritten economic story and conceptual examples, and a deterministic outcome-blind sample of 30 hypotheses per task, consisting of 10 from each primary run. Each task codebook contains six to ten substantive mechanism labels plus a residual Other label. The labels describe claims about firm behavior, accounting measurement or credibility, financing constraints, or market pricing rather than symbolic formula structures. Before coding, the codebook is frozen at SHA-256 `dc39f810c1e03c65f3b917961ec915055adc71229786d17e449008ca7bcbb2dd`. The recorded codebook hash is re-verified whenever coding outputs are validated or assembled.

Two independent GPT-5.5 coding sessions classify every record. Context is isolated by task, and each request contains only that task's frozen codebook and at most 20 blind records. Each session assigns exactly one primary mechanism and a confidence score from one to five. Exact agreement is 88.8%, with Cohen's $\kappa = 0.887$. The 1,081 disagreements are sent to fresh, task-isolated adjudication calls that observe the blind record, frozen codebook, and the two proposed labels. The assembled `mechanism_codes.csv` contains exactly one row per exported record; the residual Other category accounts for 1.88% of observations.

For the primary-run recurrence analysis, a task–mechanism–sign label is present in a run when at least one evaluated hypothesis in that task receives the mechanism label and declared direction. Pairwise Jaccard overlap is computed on these sets separately within each task and then macro-averaged equally across the 28 tasks. Horse-race-survivor overlap repeats the calculation after joining the coded records to the terminal horse-race sets on run, task, and stored proposal name. This key is one-to-one across all 9,656 coded records—no task within a run contains two evaluated hypotheses sharing a proposal name—and the normalized expression

and the provenance hash used elsewhere in the post-search analysis give the same match. The assembly verifies that the number of matched survivors equals the number of survivors in each run.

The mechanism-diversity diagnostic counts distinct substantive labels and computes

$$\mathrm{HHI}_{kr} = \sum_{m} s_{mkr}^{2},$$

where s_{mkr} is mechanism m's share of non-Other evaluated hypotheses in task k and run r. Higher values indicate greater concentration in fewer mechanisms. Other is excluded from both the count and HHI because it is a residual category rather than a coherent mechanism. Appendix Table B.5 reports task-level results for A1–A3 together with the B1–B3 median and full range.

A.9 Outcome-Blind Coding of Critic Objections

The coding corpus contains every structured critic judgment returned in B1–B3, A2, and A3; A1 has no critic. The export covers 1,400 pair-generations and yields 8,398 judgments. Each record contains only run and task provenance, generation, target name, the contemporaneous verdict and score, and the critic's rationale. It excludes the target expression and mechanism statement, the target's empirical outcome, coverage and execution fields, survivor status, and all post-search results. A rationale can quote statistics from earlier hypotheses in the visible archive, but the coding task never receives the later empirical outcome of the current target.

Before coding, we freeze a taxonomy that assigns one mutually exclusive dominant category and permits up to two nonexclusive secondary categories. The dominant categories distinguish novelty or redundancy, economic logic, measurement of the stated construct, identification or attribution of the claimed cross-theme contribution, functional-form behavior, data feasibility, failure to respond to an archived diagnostic, and judgments containing no material objection. Measurement concerns identify cases in which chosen variables, scaling, stock–flow treatment, or accounting interpretation do not implement the stated construct. Functional-form concerns instead identify behavior induced by an operator applied to otherwise relevant inputs. Archive-feedback concerns are reserved for failures to respond to an empirical diagnostic already visible within the same task, rather than simple formula similarity.

A separate novelty field distinguishes a critic-classified published rediscovery, archive du-

plication or cosmetic variation, limited incremental extension, substantive new condition or combination, and "not discussed." These are process-internal, contemporaneous classifications, not exhaustive literature verification. The critic's comparison set consists of proposals visible in the contemporaneous task archive and published signals available from its model knowledge; it is not an exhaustive literature search. The codebook is frozen at SHA-256 `a42c58ddad4c265b595d125155466bf422370f988e7e21151b6703b914fd4eb2` before any classifications are returned.

Two independent GPT-5.6 Terra sessions code the full corpus in batches of at most 20 records. Each receives the same frozen codebook but no output from the other session. For each judgment, it assigns one dominant object, up to two secondary objects, a contemporaneous novelty classification, a material-concern indicator, confidence from one to five, and a short supporting excerpt. Exact agreement is 88.3% for the dominant category, 90.0% for novelty, and 99.3% for the material-concern indicator; the corresponding Cohen's κ statistics are 0.848, 0.868, and 0.897. The 1,678 judgments with a disagreement on at least one of these three fields are sent to a third fresh session for adjudication under the same frozen codebook.

Record identifiers are validated against the export before assembly. Twelve omitted or malformed identifiers from the first coding pass and ten from the second are submitted for fresh blind recoding using only the original record and the corresponding coder-specific prompt. Format-only normalization removes redundant or excess secondary labels without changing the dominant category, novelty assessment, or material-concern indicator. The adjudicated `critique_codes.csv` contains exactly one row for each of the 8,398 judgments. Tables 9 and 10 are generated only after adjudication.

The novelty classification and flag decision are taken from the same contemporaneous critic judgment. Their association therefore describes how the critic implements its review standard; it is not independent evidence that a proposal is or is not novel.

A.10 Recorded Trajectory Evidence for the Four Case Studies

The artifact bundle preserves the producing generation for each of the 11 complete-protocol survivor records. Section 7 examines four especially legible trajectories involving Signals 1, 2, 4, 6, and 10. Each bundle contains the rendered prompts, visible model responses, final and superseded proposals, critic judgments where applicable, validation and evaluation records, the task contract, horse-race selection record, and post-search evidence. Signal identity is verified

using the immutable provenance hash employed throughout the post-search pipeline. Appendix Table B.9 reports the terminal expressions. The details below document transitions that cannot be recovered from those terminal expressions alone.

Signals 1 and 2: construct repair and representational extension. For Signal 1, the flagged proposal `buy_to_build_spending_ratio` divided five-year acquisition spending by the larger of five-year R&D spending and depreciation. The critic observed that depreciation generally determined the denominator, so the expression implemented acquisitions relative to capital consumption rather than the proposed build-versus-buy construct. The revision divides five-year acquisitions by the sum of five-year acquisitions and R&D spending. The revised share lies between zero and one when the two measured inputs are nonnegative and their sum is positive, and it compares flows measured over the same horizon.

Signal 2 appears in the next generation of the same task and represents the mechanism through a different financial-statement contrast. One component subtracts recognized intangible assets scaled by total assets from five-year cumulative R&D spending scaled by total assets. Neither accumulated R&D spending nor booked intangible assets is a pure stock of internally built or purchased capability, and the expression does not construct capitalized and amortized knowledge capital. It instead contrasts the different reporting footprints of R&D spending, which is largely expensed as incurred, and recognized intangible assets. The recorded rationale explicitly links this representation to the preceding growth-sourcing mechanism.

Signal 4: conditional reformulation. The initial proposal family measured the relative change in R&D versus capital expenditure. Critique identified the unresolved ambiguity between deliberate reallocation toward intangible investment and an inability to maintain physical investment. The terminal expression multiplies the rank of the R&D-versus-capital-expenditure change, scaled by average sales, by the rank of gross-profit growth. Gross-profit growth is an operating-performance condition, not another financing or investment input, and does not by itself establish successful innovation. The revision therefore changes an unconditional investment-tilt hypothesis into a sharper prediction about when that accounting pattern is informative.

Signal 6: corroboration through a weakest-link rule. The critic flagged the predecessor as too close to familiar profitability and cash-exclusion constructions. The revision introduced an R&D-expensing channel and changed the components from an additive score to a binding

conjunction. Its first component is

$$\frac{\texttt{oiadp} + \texttt{xrd}}{\texttt{at}},$$

with missing current R&D expense replaced by zero. This component mechanically reverses current R&D expensing but neither constructs an R&D capital stock nor applies an amortization schedule. We therefore call it R&D-expense-adjusted profitability. The second component is

$$-\frac{\texttt{act} - \texttt{che}}{\texttt{sale}},$$

which assigns a higher value to firms with lower noncash current-asset intensity relative to sales. Both components are ranked within industry before `MIN` is applied. The score therefore cannot be high unless both R&D-expense-adjusted profitability and the balance-sheet condition are strong; one component cannot compensate for weakness in the other.

Signal 10: changing a variable's functional role after archived evidence. Signal 10 comes from A1 and therefore has no critic intervention. The relevant feedback is the empirical archive supplied to the next generation. The earlier `AccrualGapUnderRolloverStress` hypothesis produces a Fama–MacBeth statistic of 0.195 but a raw Q5-minus-Q1 alpha of -7.0% with $t_\alpha = -4.627$, consistent with its declared negative direction. The subsequent proposal's recorded rationale interprets this disparity as suggesting a tail-oriented portfolio pattern rather than a monotone cross-sectional slope. Alternative explanations, including weighting, outliers, and differences between the two estimators, remain possible; the trace establishes the proposal's response to the diagnostic, not the truth of its interpretation.

The terminal expression places the income–cash-flow gap inside a nested `MIN` construction together with five-year trough cash-flow coverage and net cash. The accrual measure therefore no longer serves as a stand-alone continuous score. It becomes one binding condition in a broader rollover-resilience hypothesis.

Signals 1, 4, and 6 are revisions following critic feedback. Signal 2 is accepted without revision in the generation after Signal 1, and Signal 10 follows archived empirical evidence without a critic. These cases are selected ex post for survival and legibility and therefore do not identify the frequency, average value, or causal effect of these research moves. They document observable changes in construct measurement, economic conditioning, and functional form; they do not establish the associated mechanisms or provide direct evidence about latent cognitive

processes.

Appendix B Supplementary Tables

This appendix collects the supporting tables referenced in the main text and in Appendix A. Tables are ordered by the section of the paper whose material they support.

Table B.1: Operator Classes of the Hypothesis Language

Class (count)	Operators	Economic role
Scaling and change (3)	`RATIO(a, b)`, `GROWTH(x, lag)`, `DELTA(x, lag)`	Ratios of accounting items; year-over-year growth and level changes.
Time-series state (10)	`MA(x, k)`, `TREND(x, k)`, `VOL(x, k)`, `ACCEL(x)`, `LAG(x, d)`, `TS_MIN(x, k)`, `TS_MAX(x, k)`, `TS_SUM(x, k)`, `TS_COUNT(x, k)`, `TS_RANK(x, k)`	Within-firm smoothing, slope, volatility, acceleration, lags, distance from own extremes, cumulative flows, reporting persistence.
Composition (6)	`ADD(a, b)`, `SUB(a, b)`, `MUL(a, b)`, `SUMRANK(a, b)`, `INDICATOR(x, threshold)`, `COALESCE(a, b)`	Arithmetic interaction, rank composites, threshold indicators, economically justified missing-value fallback.
Mathematical transforms (7)	`ABS(x)`, `SIGN(x)`, `LOG(x)`, `INV(x)`, `NEG(x)`, `MIN(a, b)`, `MAX(a, b)`	Nonlinear reshaping and weakest-link or strongest-link constructions.
Cross-sectional and industry-relative (6)	`RANK(x)`, `ZSCORE(x)`, `WINSOR(x, p)`, `IND_ADJ(x)`, `IND_ZSCORE(x)`, `IND_RANK(x)`	Cross-sectional ranking, standardization, winsorization; industry-median adjustment and industry-relative scores.

The language contains 32 operators with explicit arity and parameter domains. Arguments a, b, and x denote nested sub-expressions or variables; named arguments are integer parameters (lag, window, threshold, or tail percentile). Operators compose recursively, so the admissible space contains ratios of flows and stocks, temporal states, industry-relative measures, and rank composites of any of these.

Table B.2: Cross-Theme Tasks

Theme pair	Economic objective supplied to the task
Profitability × Valuation	Highly profitable firms that are also cheap combine strong fundamentals with low market expectations.
Profitability × Investment	Profitable firms that do not invest aggressively can be underpriced cash generators, while profitable firms that invest aggressively can be overpriced because the market credits them for both profitability and growth.
Profitability × Accruals	Profitable firms are more credible when profits are backed by cash rather than accrual accounting.
Profitability × Quality	High profitability is more valuable when it is stable, persistent, and financially healthy.
Profitability × Financing	Profitable firms that return capital may reveal underpriced cash generation, while profitable firms issuing capital may reveal overvaluation or funding needs.
Profitability × Intangibles	Intangible-intensive firms with current profitability may be underpriced because accounting understates their productive asset base.
Profitability × Distress	Profitability may be mispriced differently for firms near financial distress.
Valuation × Investment	Cheap firms that also avoid aggressive investment can be underappreciated disciplined capital allocators, while expensive aggressive investors may be overvalued growth stories.
Valuation × Accruals	Cheap firms can be value traps when reported earnings rely on accruals rather than cash.
Valuation × Quality	Cheap firms with high earnings quality are more likely to be mispriced than cheap firms whose low valuation reflects distress.
Valuation × Financing	Cheap firms where management repurchases shares may reveal stronger undervaluation, while expensive firms issuing equity may reveal overvaluation.
Valuation × Intangibles	Accounting can understate book value for intangible-intensive firms, making standard valuation ratios noisy.
Valuation × Distress	Cheap stocks can be compensated value opportunities or distressed lottery-like traps.
Investment × Accruals	Asset growth driven by accruals can reverse differently from growth driven by real investment.
Investment × Quality	Investment is more informative when paired with quality: disciplined investment by high-quality firms may differ from asset growth by fragile firms.
Investment × Financing	Aggressive investment funded by external financing can reveal overvaluation, market timing, or empire building.
Investment × Intangibles	Tangible and intangible investment can carry different return implications.
Investment × Distress	Aggressive investment by distressed firms may indicate value-destroying expansion or urgent balance-sheet strain, while disciplined investment may signal resilience.
Accruals × Quality	Low accruals combined with persistent, stable earnings should identify genuinely cash-backed reported performance.
Accruals × Financing	External financing combined with aggressive accruals can indicate overvaluation, earnings management, or balance-sheet pressure.
Accruals × Intangibles	Intangible-intensive firms can have accounting noise from both expense treatment and accrual components.
Accruals × Distress	Accrual problems can be more severe for financially fragile firms, where earnings management and liquidity pressure are more likely.
Quality × Financing	Financing actions are more informative when paired with firm quality.

Continued on next page

Table B.2 continued

Theme pair	Economic objective supplied to the task
Quality × Intangibles	Intangible investment is more credible when paired with stable cash flows, earnings persistence, and balance-sheet quality.
Quality × Distress	Quality and distress are opposing but not identical dimensions.
Financing × Intangibles	External financing for intangible investment can indicate valuable innovation funding or market-timed issuance.
Financing × Distress	Financing behavior has different meaning for distressed firms.
Intangibles × Distress	Intangible-intensive firms can look distressed under traditional accounting even when they own productive knowledge capital.

All $\binom{8}{2} = 28$ pairwise combinations of the eight themes. The objective shown is the first sentence of each task's economic story; the task also carries example interaction ideas, theme-specific variable menus, and preferred operators, which are conceptual anchors rather than templates. Each proposal must use at least one variable exclusive to each theme of the pair.

Table B.3: Primary Discovery Yield by Theme Pair

Theme pair	B1 hit	B2 hit	B3 hit	Median [range]	Survivors median [range]
Accruals x Distress	18.9	35.8	72.2	35.8 [18.9, 72.2]	1 [1, 1]
Accruals x Financing	63.3	59.3	53.4	59.3 [53.4, 63.3]	3 [2, 4]
Accruals x Intangibles	46.7	53.3	55.0	53.3 [46.7, 55.0]	3 [3, 4]
Accruals x Quality	66.7	53.8	94.5	66.7 [53.8, 94.5]	3 [1, 3]
Financing x Distress	33.9	31.0	23.7	31.0 [23.7, 33.9]	2 [2, 3]
Financing x Intangibles	48.3	53.3	35.6	48.3 [35.6, 53.3]	4 [3, 7]
Intangibles x Distress	56.7	37.3	30.5	37.3 [30.5, 56.7]	3 [3, 3]
Investment x Accruals	57.6	70.7	64.9	64.9 [57.6, 70.7]	4 [3, 5]
Investment x Distress	43.9	50.0	42.4	43.9 [42.4, 50.0]	2 [1, 3]
Investment x Financing	47.4	56.9	86.4	56.9 [47.4, 86.4]	4 [3, 6]
Investment x Intangibles	62.5	57.9	51.7	57.9 [51.7, 62.5]	4 [2, 5]
Investment x Quality	62.1	57.9	63.2	62.1 [57.9, 63.2]	3 [1, 5]
Profitability x Accruals	80.0	80.4	84.9	80.4 [80.0, 84.9]	3 [3, 4]
Profitability x Distress	50.8	51.8	50.0	50.8 [50.0, 51.8]	2 [2, 5]
Profitability x Financing	44.1	62.7	57.9	57.9 [44.1, 62.7]	3 [1, 4]
Profitability x Intangibles	77.8	80.7	85.7	80.7 [77.8, 85.7]	2 [2, 3]
Profitability x Investment	71.7	72.9	65.5	71.7 [65.5, 72.9]	2 [2, 3]
Profitability x Quality	78.6	80.0	80.4	80.0 [78.6, 80.4]	1 [1, 2]
Profitability x Valuation	87.9	60.3	89.8	87.9 [60.3, 89.8]	3 [3, 3]
Quality x Distress	57.4	11.3	44.2	44.2 [11.3, 57.4]	2 [1, 5]
Quality x Financing	69.5	78.0	27.6	69.5 [27.6, 78.0]	2 [1, 3]
Quality x Intangibles	62.1	72.9	67.8	67.8 [62.1, 72.9]	3 [2, 4]
Valuation x Accruals	57.9	74.1	67.9	67.9 [57.9, 74.1]	3 [3, 4]
Valuation x Distress	19.3	36.8	46.4	36.8 [19.3, 46.4]	2 [1, 2]
Valuation x Financing	50.8	74.1	78.0	74.1 [50.8, 78.0]	2 [1, 2]
Valuation x Intangibles	96.7	90.0	88.3	90.0 [88.3, 96.7]	3 [2, 5]
Valuation x Investment	56.1	93.1	67.8	67.8 [56.1, 93.1]	2 [1, 4]
Valuation x Quality	83.6	35.7	71.9	71.9 [35.7, 83.6]	4 [1, 4]

Hit rates are percentages of evaluated proposals that pass the declared-sign Fama–MacBeth screen. Brackets give the full B1–B3 range; each primary run is kept separate. Survivor counts come from the final within-pair horse race.

Table B.4: Theme-Level Marginal Discovery Yield in the Primary Runs

Theme	B1 hit	B2 hit	B3 hit	Median [range]	Survivors per task, median [range]
Profitability	69.8	69.8	73.3	69.8 [69.8, 73.3]	2.57 [2.14, 3.00]
Valuation	64.8	66.6	73.2	66.6 [64.8, 73.2]	2.57 [2.43, 2.86]
Quality	68.5	56.3	64.2	64.2 [56.3, 68.5]	2.29 [2.14, 3.00]
Intangibles	64.2	63.6	59.1	63.6 [59.1, 64.2]	3.57 [2.71, 3.71]
Investment	57.4	65.7	63.1	63.1 [57.4, 65.7]	2.86 [2.71, 3.71]
Accruals	56.0	61.2	70.0	61.2 [56.0, 70.0]	2.86 [2.57, 3.29]
Financing	51.2	59.4	51.8	51.8 [51.2, 59.4]	3.00 [2.86, 3.00]
Distress	40.4	36.5	43.8	40.4 [36.5, 43.8]	2.00 [2.00, 2.71]

Each theme participates in seven tasks. The hit rate pools the evaluated proposals of those seven tasks within a run; survivors per task average the final horse-race survivors over the same seven tasks. Runs are kept separate.

Table B.5: Pair-Level Mechanism Breadth and Concentration

	Distinct mechanism count				HHI			
Theme pair	B median [range]	A1	A2	A3	B median [range]	A1	A2	A3
Accruals x Distress	8 [8, 8]	7	7	8	0.204 [0.189, 0.324]	0.596	0.234	0.162
Accruals x Financing	7 [6, 7]	5	7	7	0.238 [0.213, 0.238]	0.270	0.363	0.290
Accruals x Intangibles	8 [8, 8]	7	8	8	0.152 [0.148, 0.207]	0.250	0.173	0.146
Accruals x Quality	7 [7, 8]	6	6	7	0.176 [0.169, 0.219]	0.337	0.233	0.185
Financing x Distress	8 [7, 8]	7	7	8	0.204 [0.194, 0.218]	0.477	0.228	0.217
Financing x Intangibles	8 [7, 8]	6	6	7	0.254 [0.243, 0.311]	0.448	0.240	0.291
Intangibles x Distress	9 [9, 9]	8	9	9	0.136 [0.128, 0.150]	0.199	0.150	0.147
Investment x Accruals	9 [7, 9]	6	8	7	0.249 [0.223, 0.297]	0.301	0.185	0.184
Investment x Distress	9 [8, 9]	9	8	9	0.195 [0.158, 0.248]	0.342	0.164	0.165
Investment x Financing	8 [8, 8]	6	8	8	0.189 [0.187, 0.247]	0.288	0.198	0.172
Investment x Intangibles	8 [8, 8]	8	8	8	0.228 [0.194, 0.229]	0.295	0.175	0.184
Investment x Quality	7 [5, 8]	7	7	6	0.213 [0.181, 0.453]	0.271	0.192	0.257
Profitability x Accruals	8 [8, 8]	4	8	7	0.213 [0.207, 0.243]	0.542	0.213	0.205
Profitability x Distress	7 [6, 8]	8	6	6	0.235 [0.195, 0.279]	0.474	0.194	0.204
Profitability x Financing	6 [6, 8]	7	7	8	0.286 [0.203, 0.287]	0.324	0.207	0.234
Profitability x Intangibles	7 [6, 8]	7	6	8	0.193 [0.168, 0.277]	0.241	0.274	0.307
Profitability x Investment	8 [7, 8]	8	7	8	0.248 [0.220, 0.346]	0.212	0.170	0.153
Profitability x Quality	8 [7, 8]	5	8	6	0.193 [0.158, 0.198]	0.494	0.192	0.289
Profitability x Valuation	8 [6, 8]	7	8	8	0.230 [0.220, 0.235]	0.331	0.151	0.163
Quality x Distress	8 [8, 8]	8	8	7	0.177 [0.148, 0.182]	0.191	0.231	0.259
Quality x Financing	7 [6, 8]	8	6	7	0.316 [0.278, 0.334]	0.505	0.375	0.162
Quality x Intangibles	8 [8, 8]	5	7	8	0.244 [0.150, 0.306]	0.288	0.226	0.172
Valuation x Accruals	8 [7, 8]	7	6	8	0.193 [0.168, 0.258]	0.267	0.277	0.158
Valuation x Distress	8 [8, 8]	8	7	8	0.209 [0.142, 0.320]	0.193	0.231	0.194
Valuation x Financing	7 [6, 8]	6	8	7	0.226 [0.205, 0.473]	0.369	0.221	0.346
Valuation x Intangibles	8 [7, 8]	7	6	5	0.179 [0.158, 0.209]	0.213	0.264	0.372
Valuation x Investment	8 [7, 8]	8	8	8	0.222 [0.203, 0.222]	0.380	0.150	0.259
Valuation x Quality	8 [8, 8]	6	8	8	0.210 [0.173, 0.260]	0.246	0.254	0.190

B summarizes the median and full range across B1–B3 without pooling records. A1–A3 are the three single-run configuration comparisons. Counts and HHI are computed within a run and task from evaluated hypotheses, excluding the residual Other label; lower HHI indicates less concentration. A1 is the direct no-critique contrast because it retains the primary proposer. A2–A3 instead change proposer identity and are descriptive model contrasts.

Table B.6: Expressions of the Selected Primary-Run Survivors

Run	Signal	Sign	Expression
B1	Symmetric Policy And Net Issuance Composite	negative	`SUMRANK(NEG(SUB(RATIO(dp, at), MA(RATIO(dp, at), 5))), RATIO(SUB(sstk, prstkc), at))`
B1	Durable Profit Per Hard Dollar Value	positive	`SUMRANK(RATIO(SUB(ceq, COALESCE(intan, 0)), MUL(prcc_f, csho)), SUMRANK( TS_MIN(RATIO(ADD(COALESCE(xrd, 0), COALESCE(xad, 0)), at), 5), TS_MIN(RATIO( SUB(sale, cogs), SUB(at, COALESCE(intan, 0))), 5)))`
B1	Burning A War Chest Raised Earlier	negative	`MUL(WINSOR(RATIO(SUB(dvt, oancf), at), 1), RANK(RATIO(TS_SUM(sstk, 3), at)))`
B2	Industry Adjusted Own Peak Deferral With Burn	negative	`MIN(RANK(NEG(IND_ADJ(RATIO(RATIO(dp, MAX(SUB(at, act), dp)), TS_MAX(RATIO(dp, MAX(SUB(at, act), dp)), 5))))), RANK(RATIO(SUB(ADD(DELTA(ADD(dltt, dlc), 1), SUB(DELTA(ceq, 1), SUB(ib, dvc))), DELTA(che, 1)), at)))`
B2	Pre Investment Cash Plus Composition Rank	positive	`SUMRANK(RATIO(ADD(oancf, ADD(COALESCE(xrd, 0), COALESCE(xad, 0))), at), RATIO(SUB(TS_SUM(ADD(COALESCE(xrd, 0), COALESCE(xad, 0)), 5), COALESCE(intan, 0)), ADD(TS_SUM(ADD(COALESCE(xrd, 0), COALESCE(xad, 0)), 5), COALESCE(intan, 0))))`
B2	Total Funding Requirement Burden	negative	`RANK(RATIO(SUB(ADD(capx, dlc), oancf), at))`
B3	Normalized firm level yield with wide mix	positive	`MUL(IND_RANK(RATIO(MA(ebitda, 3), ADD(MUL(prcc_f, csho), ADD(dltt, dlc)))), RANK(COALESCE(RATIO(SUB(prstkc, dvt), ADD(prstkc, dvt)), SIGN(SUB(prstkc, dvt)))))`
B3	Distributions exceeding internal cash generation before maturities	negative	`MUL(RANK(RATIO(SUB(TS_SUM(dvt, 3), TS_SUM(oancf, 3)), at)), RANK(RATIO(dlc, act)))`
B3	Liquid Assets Plus Normalized Earnings Yield Composite	positive	`SUMRANK(RATIO(che, at), RATIO(MA(oiadp, 5), MUL(prcc_f, csho)))`

Expressions are reproduced exactly as proposed and evaluated. "Sign" is the direction declared by the proposer before evaluation.

Table B.7: Model-Call Accounting by Run

Run	Calls	Proposer calls	Critic calls	Revision calls	Output tokens (m)	Flagged (%)
B1	840	560	280	280	4.32	49.3
B2	840	560	280	280	4.32	52.7
B3	839	559	280	279	4.37	51.4
A1	280	280	0	0	1.76	—
A2	840	560	280	280	4.37	64.5
A3	840	560	280	280	3.72	59.4

Counts come from recorded model responses. Proposer calls include one initial call per pair-generation and any revision calls; revision calls are therefore proposer calls minus 280. The flagged share divides critic flags by the 1,680 requested proposal slots. Output-token counts are recorded provider usage and are reported as process quantities rather than equal-compute controls.

Table B.8: Run-Level Multiplicity Diagnostics

		Threshold on $\lvert t \rvert$		Rejections		
Run	M_r	Bonferroni, Holm	BY	Bonferroni	Holm	BY
B1	1,603	4.165	3.071	230	235	544
B2	1,599	4.164	3.053	250	264	576
B3	1,607	4.165	3.012	343	358	663
A1	1,610	4.166	2.929	433	457	873
A2	1,612	4.166	3.315	118	120	235
A3	1,614	4.166	3.192	163	166	363

Each run is a separate testing family, and M_r is its number of usable full-sample Fama–MacBeth statistics. All three procedures operate at $q = \alpha = 0.05$ on the two-sided reference p-values defined in Appendix A.6. The Bonferroni and Holm columns share a threshold because the two procedures coincide at their most conservative step. The BY column reports the threshold implied by the largest index the procedure rejects, which depends on the realized p-value distribution and therefore varies across runs. Bonferroni and Holm are reported as family-wise-error diagnostics; only BY enters the cumulative protocol. None of the three conditions on the declared direction.

Table B.9: Expressions of the Full-Pipeline Signals

ID	Run	Signal	Sign	Expression
S1	B1	purchased share of growth spending	positive	SUB(RANK(MA(RATIO(oancf, at), 3)), RANK(RATIO(TS_SUM(COALESCE(aqc, SUB(at, at)), 5), ADD(TS_SUM(COALESCE(aqc, SUB(at, at)), 5), TS_SUM(COALESCE(xrd, SUB(at, at)), 5)))))
S2	B1	knowledge capital built versus bought	positive	SUMRANK(MA(RATIO(oancf, at), 3), WINSOR(SUB(RATIO(TS_SUM(COALESCE(xrd, SUB(at, at)), 5), at), RATIO(COALESCE(intan, SUB(at, at)), at)), 1))
S3	B2	Predictable Accruals Sustained Profit	positive	MUL(IND_RANK(NEG(WINSOR(VOL(RATIO(DELTA(SUB(wcap, che), 1), at), 5), 1))), IND_RANK(WINSOR(MA(RATIO(SUB(revt, cogs), at), 3), 1)))
S4	B2	Funded Research Tilt With Profit Delivery	positive	MUL(RANK(NEG(RATIO(SUB(DELTA(capx, 1), DELTA(COALESCE(xrd, 0), 1)), MA(sale, 3)))), RANK(GROWTH(SUB(sale, cogs), 1)))
S5	B3	Working Capital Bloat Vs Peers	positive	SUMRANK(NEG(IND_ZSCORE(RATIO(ADD(SUB(wcap, che), dlc), sale))), IND_ZSCORE(RATIO(SUB(revt, cogs), at)))
S6	B3	Conservatism Adjusted Profit Lean Claims	positive	MIN(IND_RANK(RATIO(ADD(oiadp, COALESCE(xrd, 0)), at)), IND_RANK(NEG(RATIO(SUB(act, che), sale))))
S7	A1	Sustained Coverage With Capped Cushion	positive	WINSOR(MA(RATIO(ADD(SUB(revt, cogs), MIN(ADD(xsga, COALESCE(xrd, 0)), ADD(ADD(xint, txt), dlc))), ADD(ADD(xint, txt), dlc)), 3), 1)
S8	A1	Broad Universe Profitable Value Model	positive	SUMRANK(RATIO(ADD(SUB(revt, ADD(cogs, COALESCE(xsga, 0))), COALESCE(xrd, 0)), at), SUMRANK(NEG(RATIO(SUB(ib, COALESCE(oancf, ADD(ib, dp))), at)), SUMRANK(RATIO(ADD(ceq, TS_SUM(COALESCE(xrd, 0), 3)), MUL(prcc_f, csho)), NEG(COALESCE(RATIO(SUB(ADD(dltt, dlc), che), oibdp), RATIO(SUB(ADD(dltt, dlc), che), at))))))
S9	A1	Negative Working Capital Profit Sort	positive	MUL(INDICATOR(NEG(wcap), 0), IND_ZSCORE(RATIO(SUB(revt, cogs), at)))
S10	A1	Trough Coverage Clean Accrual Rollover	positive	MIN(IND_RANK(TS_MIN(RATIO(SUB(oancf, dp), lt), 5)), MIN(IND_RANK(RATIO(SUB(che, dlc), at)), IND_RANK(NEG(RATIO(SUB(pi, oancf), revt)))))
S11	A2	Gross Margin Inventory Efficiency	positive	MUL(IND_RANK(RATIO(SUB(revt, cogs), sale)), IND_RANK(NEG(RATIO(invt, sale))))

Expressions are reproduced exactly from the final evaluated proposals. Signal IDs S1–S11 match Table 8 and the trajectory case studies in Section 7.